اللجنة الأكاديمية

Team ZerOne
Computer & Networks Engineers

دفتر:

احصاء واحتمالات

الفصل الأول (2021)

هندسة الحاسوب والشبكات - البوليتكتنك

wed

13.Oct.2021

Probability

Ch.1 : probability

Sec. 1.1 : probability Definition.

- Elements of probability: أساسيات/عناصر الاحتمالات

1) Sample space - الفضاء العينيّ

The set of all expected outcomes resulting from the probability experiment denoted by Ω. يرمز إليها

مجموعة كل النتائج / الاحتمالات الممكنة

2) Event - الحادث

Any subset of Ω denoted by any letter (A, B, C, ...). مجموعة جزئية

$A \subseteq \Omega$ ← subset

$B \not\subseteq \Omega$ ← not subset

$\Omega = \{1, 2, 3, 4\}$

$A = \{1, 3\}$

$B = \{0, 5\}$

Ex- In the experiment of [tossing a coin] [one time]. رمي قطعة نقد — مرة واحدة

Find Ω.

H: Head صورة

T: Tail كتابة

Sol: $\Omega = \{H, T\}$

Ex- In the experiment of tossing a coin two times.

Find Ω.

Sol: $\Omega = \{HH, HT, TH, TT\}$ ← two times or tossing two coin the same

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prob.

[3] prob. of Event - احتمال الحادث

The prob. of event A is defined by:

$P(A) = \frac{\text{The number of elements in } A}{\text{The number of elements in } \Omega}$

$\{\} / \phi \subseteq \Omega$ (subset)

Three Types of Event:
- Impossible → $P(A) = 0$ → %0
- sure → $P(A) = 1$ → %100
- ordinary → $0 < P(A) < 1$

Ex- In the experiment of tossing a coin two times.
Find: [1] The prob. of getting only one head.
[2] The prob. of getting at least one head.

Sol: $\Omega = \{HH, HT, TH, TT\}$

[1] $A = \{HT, TH\}$, $P(A) = 2/4 = 1/2$

[2] $B = \{HT, TH, HH\}$, $P(B) = 3/4$

Ex- In the experiment of tossing a coin three times.
Find: [1] The prob. of getting only two head.
[2] The prob. of getting at least two heads.
[3] The prob. of getting at most one head.
[4] The prob. of getting no head.

Sol:

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$\Omega = \{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \}$ $n(\Omega) = 8$

[1] $A = \{ HHT, HTH, THH \}$, $P(A) = 3/8$ [only two heads]

[2] $B = \{ HHH, HHT, HTH, THH \}$, $P(B) = 4/8 = \boxed{1/2}$ [at least two heads]

[3] $C = \{ HTT, THT, TTH, TTT \}$, $P(C) = 4/8 = \boxed{1/2}$ [at most one head]

[4] $D = \{ TTT \}$; $P(D) = 1/8$ [no head]

Ex- In the experiment of the [rolling (رمي حجر نرد) one dice], Find The prob. [1] getting an odd number.

[2] getting an aprime number.

Sol: $\Omega = \{ 1, 2, 3, 4, 5, 6 \}$

[1] $A = \{ 1, 3, 5 \} \rightarrow P(A) = 3/6 = \boxed{1/2}$

[2] $B = \{ 1, 2, 3, 5 \} \rightarrow P(B) = 4/6 = \boxed{2/3}$

Ex- In the experiment of rolling two dices, Find The prob. of : [1] The sum of faces in nine.

[2] one of the faces is twice the other.

[3] The second entery is triple the first.

[4] The first entery (element) is prime.

[5] The sum of the faces is a prime less than 4.

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Soh: $S = \{(1,1), (1,2), (1,3) \dots (1,6),$
$(2,1), (2,2), (2,3) \dots (2,6),$
$(3,1), (3,2), (3,3) \dots (3,6),$ $n(S) = 36$
$(4,1), (4,2) \dots (4,6),$
$(5,1), (5,2) \dots (5,6),$
$(6,1), (6,2), \dots (6,6)\}$

[1] $A = \{(3,6), (6,3), (4,5), (5,4)\}$, $P(A) = 4/36 = 1/9$

[2] $B = \{(2,4), (4,2), (3,6), (6,3), (1,2), (2,1)\}$, $(x, 2x)$, $(2x, x)$
$P(B) = 6/36 = 1/6$

[3] $C = \{(1,3), (2,6)\}$, $P(C) = 2/36 = 1/18$ $(x, 3x)$

[4] $D = \{(1,1) \dots (1,6), (2,1) \dots (2,6), (3,1) \dots (3,6), (5,1) \dots (5,6)\}$
$P(D) = 24/36 = 2/3$

[5] $E = \{(1,1), (1,2), (2,1)\}$, $P(E) = 3/36 = 1/12$

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Sec. 1.2: Set Operations

■ prob. Rules:

1. $P(\emptyset) = 0$, $P(\Omega) = 1$
2. $0 \leq P(A) \leq 1$
3. If A and B are independant events, then:

$P(A \cap B) = P(A) \cdot P(B)$ ← التقاطع

when. $A \cap B$ = A intersects B

= A and B

= both of A and B

* Definition: ① The two events are independant (مستقلين), if there is no relation between them and they come from the same background.

② The two events are disjoint (منفصلين), if there is no relation between them and they come from different background.

4. If A and B are disjoint events, then:

$P(A \cap B) = 0$, $A \cap B = \emptyset$

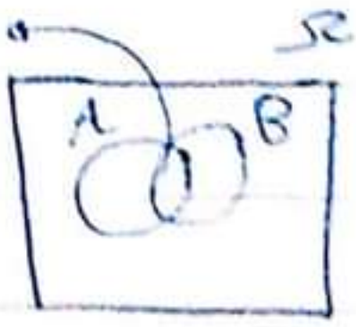
* Remark: If A and B are independant, then:

$P(A \cap B) \neq 0$,

5. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

where $A \cup B$ = A unionB = A orB

= At Least one of them or both.



[6] $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) - P(A \cap B \cap C)$.

ote: [1] $A \cap B = B \cap A$

[2] $A \cap A = A$

[3] $(A \cup B) \cap C = (A \cup C) \cap (A \cup B)$

[4] $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof [6]:

Let $D = B \cup C$

$P(A \cup B \cup C) = P(A \cup D)$

$= P(A) + P(D) - P(A \cap D)$

$= P(A) + P(B \cup C) - P(A \cap (B \cup C))$

$= P(A) + P(B) + P(C) - P(B \cap C) - P((A \cap B) \cup (A \cap C))$

Let $E = A \cap B$ and $F = A \cap C$

$P((A \cap B) \cup (A \cap C)) = P(E \cup F)$

$= P(E) + P(F) - P(E \cap F)$

$= P(A \cap B) + P(A \cap C) - P((A \cap B) \cap (A \cap C))$.

$= P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)$.

$= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C)$.

Ex- Two fire detectors A and B are working independently such that $P(A) = 0.2$ and $P(B) = 0.4$, Find the prob. of:

[1] both of them will work.

[2] at least one of them will work.

Sol: [1] $P(A \cap B) = P(A) \cdot P(B) = 0.2 * 0.4 = 0.08$.

[2] $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.2 + 0.4 - 0.08 = 0.52$

7 Complement of probability:

$\bar{A}$ = The complement of A ⇒ An elements in Ω except A.

= Ω / A , $\Omega - B$

⇒ Remarks:

1. $A \cup \bar{A} = \Omega$ 2. $A \cap \bar{A} = \phi$.

$P(\bar{A}) = 1 - P(A)$ / $P(A) = 1 - P(\bar{A})$ Rule.

* Proof [7] : since $A \cup \bar{A} = \Omega$, then

$P(A \cup \bar{A}) = P(\Omega)$

$P(A) + P(\bar{A}) - P(A \cap \bar{A}) = 1$

$P(A) + P(\bar{A}) - P(\phi) = 1$ ⇒ $\boxed{P(\bar{A}) = 1 - P(A)}$

0

* Theorem:

If A and B are independent events, then:

1. A and $\bar{B}$ are independent.
2. $\bar{A}$ and B are independent.
3. $\bar{A}$ and $\bar{B}$ are independent.

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[8] Demorgan's laws

[1] $\overline{A \cup B} = \bar{A} \cap \bar{B}$

[2] $\overline{A \cup B \cup C} = \bar{A} \cap \bar{B} \cap \bar{C}$

[3] $\overline{A \cap B} = \bar{A} \cup \bar{B}$

[4] $\overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$

Ex- A and B are independent components Items. such that $p(A \text{ works}) = 0.3$ and $p(B \text{ works}) = 0.6$. Find the prob that: [1] B will fail.

[2] at least one of them will fail.

[3] A with fail and B with fail.

[4] Both of them will fail.

Sol: [1] $P(\bar{B}) = 1 - P(B) = 1 - 0.6 = 0.4$

[2] $P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B})$

$= 1 - P(A \cap B)$

$= 1 - P(A).P(B) = 1 - (0.3 * 0.6) = 0.82$

[3] $P(A \cap \bar{B})$ ⇒ since A and B are indep.

A & $\bar{B}$

$P(A \cap \bar{B}) = P(A) \cdot P(\bar{B})$

$= (1 - P(A)) \cdot (1 - P(B))$

$= (1 - 0.3) \cdot (1 - 0.6) = 0.28$

[4] $P(\overline{A \cap B}) = 1 - P(A \cap B)$

$= 1 - P(A).P(B)$

$= 1 - (0.3 * 0.6) = 0.82$

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Proof Theorem:

$P(\bar{A} \cap \bar{B}) \overset{?}{=} P(\bar{A}) \cdot P(\bar{B})$.

$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$

$= 1 - P(A \cup B)$

$= 1 - P(A) + P(B) - P(A \cap B)$.

$= 1 - P(A) + P(B) - P(A) \cdot P(B)$.

$= P(\bar{A}) - P(B)(1 - P(A))$

$= P(\bar{A}) - P(B) \cdot P(\bar{A})$

$= P(\bar{A})(1 - P(B) = P(\bar{A}) \cdot P(\bar{B}) = P(\bar{A} \cap \bar{B})$ #.

☐ Exception law prob.

[a] $P(A - B) \equiv P(A \cap \bar{B})$

$P(A) - P(A \cap B)$.

[b] $P(\bar{A} \cap B) = P(B - A) = P(B \cap \bar{A})$

$= P(B) - P(A \cap B)$.

Theorem a & b

Ex- If prob: $P(A) = 0.2$, $P(\bar{A} \cap B) = 0.3$. Find $P(A \cup B)$.

Sol: $P(A \cup B) = P(A) + [P(B) - P(A \cap B)]$ ← $P(\bar{A} \cap B)$

$= 0.2 + 0.3 = 0.5$

$P(B) - P(A \cap B) = P(\bar{A} \cap B)$.

Ex- If $P(A) = 0.4$ and $P(A \cap \bar{B}) = 0.1$. Find

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$P(\bar{A} \cup \bar{B})$:

Sol: $P(A \cap \bar{B}) = P(A) - P(A \cap B)$

$0.1 = 0.4 - P(A \cap B)$

$P(A \cap B) = 0.3$

$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B})$

$= 1 - P(A \cap B)$

$= 1 - 0.3 = 0.7$.

Ex- If $P(A \cup B) = 0.9$, $P(A) = 0.7$ and $P(B) = 0.4$, Find: $P(\overline{A \cup B})$

Sol: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.9 = 0.7 + 0.4 - P(A \cap B)$

$P(A \cap B) = 0.2$

$P(\overline{A \cup B}) = P(\bar{A} \cap \bar{\bar{B}})$

$= P(\bar{A} \cap B) = P(B) - P(A \cap B)$

$= 0.4 - 0.2 = 0.2$

Remark : If A and B are dep. then:

$P(A \cap B) \neq 0$.

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Prob. CH.1

☐ Conditional Prob. الاحتمال المشروط

$P(A|B)$ ← given

$P(A|B) = \frac{P(A \cap B)}{P(B)}$, where

لـ احتمال حدوث

$P(A|B)$ = The prob $(A|B)$
= The prob of event A such that event B is Done.

Remarks:

① $P(\phi|A) = \frac{P(\phi \cap A)}{P(A)} = \frac{P(\phi)}{P(A)} = \frac{0}{P(A)} = 0$ ← $0 = \phi$ أي اشي لتقاطع

② $P(\Omega|A) = \frac{P(\Omega \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1$

③ $P(\bar{A}|B) = \frac{P(\bar{A} \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)}$

$= \frac{P(B)}{P(B)} - \frac{P(A \cap B)}{P(B)}$

$= 1 - P(A|B)$

④ If A and B are indep. events, then:

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

$= \frac{P(A) \cdot P(B)}{P(B)} = P(A)$.

Ex- In some population, it's found that $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$,

where A denotes to the hypertenstion, B denotes to the over wieghts. If a person chosen at random. find the prob. that he has a hypertetion, such that he has over weight.

Sol: $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = 0.4$

التباديل

Permutations : The number of ways of chosing r elements out of n elements such that the order is important, which is denoted by:

$$nP_r = \frac{n!}{(n-r)!}$$

التوافيق

Combinations : The nuber of ways of chosing r elements out of n elements such that the order is not important, which is denoted by:

$$nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Ex. How many Three digits car plates you form from {0, 1, 2, ... 9} without duplicate any number.

No. of Plates $= (10)(9)(8) = 720$ (OR)

Sol:

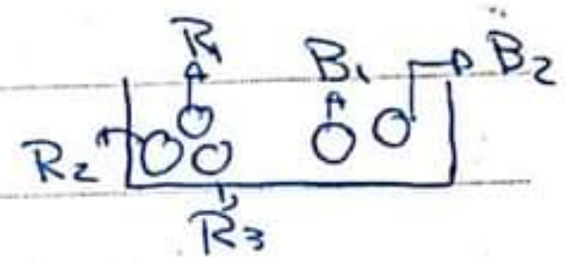
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$$10P_r = \frac{10!}{(10-3)!} = \frac{10!}{7!} = \frac{10\times9\times8\times\cancel{7!}}{\cancel{7!}} = 720$$

Ex. In the experiment of drawing two balls from a box with 3 red and 2 black balls. Find the:

1. number of ways of chosing one red ball.
2. number of ways of chosing two red balls.
3. number of ways of chosing two balls.
4. prob. of chosing one red ball.
5. prob of chosing two red balls.



Sol: 1. $n(\Omega) = \binom{5}{2} = \frac{5!}{2!(5-2)!} = \frac{5\times4\times\cancel{3!}}{2!\cdot\cancel{3!}} = 10$

$\Omega = \{R_1R_2, R_1R_3, R_2R_3, R_1B_1, R_1B_2, R_2B_1, R_2B_2, R_3B_1, R_3B_2, B_1B_2\}$.

$\binom{n}{1} = n$

$\binom{n}{0} = 1$

2. $\binom{3}{1}\binom{2}{1} = 3\times2 = 6$
3. $\binom{3}{2}\binom{2}{0} = 3\times1 = 3$
4. $n(\Omega) = \binom{5}{2} = \boxed{10}$

4. $P(1\text{ red}) = \frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}} = \frac{6}{10} = 3/5$

5. $P(2\text{ red}) = \frac{\binom{3}{2}\binom{2}{0}}{10} = \frac{3}{10}$

←

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Prob. CH.1

Ex- 10 students

5 science 3 Engineer 2. Medical.

we choose 3 students randomly.

[1] Find the prob. of chosing 2 science students.

[2] Find the prob. of chosing at least 2 eng. students.

[3] Find the prob. of chosing one science and at least one eng.

[4] Find the prob. of chosing at least one medical student.

Sol: [1] $P(2 \text{ science}) = \frac{\binom{5}{2}\binom{5}{1}}{\binom{10}{3}} = \frac{50}{120} = 5/12.$

[2] $P(\text{at least 2 eng}) = P(2 \text{ eng}) + P(3 \text{ eng})$

$= \frac{\binom{3}{2}\binom{7}{1}}{\binom{10}{3}} + \frac{\binom{3}{3}\binom{7}{0}}{\binom{10}{3}}$

$n(S) = \binom{10}{3} = \frac{10!}{3!(7!)} = \boxed{120}$

[3] $P(1 \text{ science} \cap \text{at least } 1 \text{ eng}) =$

$P(1 \text{ science} \cap 1 \text{ Eng}) + P(1 \text{ science} \cap 2 \text{ eng}) =$

$= \frac{\binom{5}{1}\binom{3}{1}\binom{2}{1}}{\binom{10}{3}} + \frac{\binom{5}{1}\binom{3}{2}\binom{2}{0}}{\binom{10}{3}}$

[4] $P(\text{at Least 1 medical}) = P(1 \text{ medical}) + P(2 \text{ medical}).$

$= \frac{\binom{2}{1}\binom{8}{2}}{\binom{10}{3}} + \frac{\binom{2}{2}\binom{8}{1}}{\binom{10}{3}}$

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prob. CH.2

CH.2 : Univariate Random variables

المتغيرات العشوائية الأحادية.

* Def : The random variable (r.v) X is a function defined from Ω to R ($X : \Omega \to R$).

domain ← Range

* Remarks :

① The range of the r.v X is denoted by (R_X), which is the set of all possible values of the r.v.

② The random variables have two main types :

[a] Discrete r.v → It takes values as distinct numbers. (أرقام منفصلة)

[b] Continuous r.v → It takes values as intervals. (فترات)

Ex : y : The number of heads in the experiment of tossing a coin 5 times.

→ $R_Y = \{0, 1, 2, 3, 4, 5\}$ Discrete r.v

Z : The weight of boxing championship.

$R_Z = \{(49, 56), (56, 63), \ldots (91, 98)\}$. Cont r.v

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prob. CH. 2

• Def : The prob. density function (P.d.f).
اقتران الكثافة الاحتمالية

If the r.v is discrete
↓
Denoted by P(X) or P(X=x) and satisfies the following:

[1] $P(X) \geq 0$, $\forall x \in R_x$

[2] $\sum_{X \in R_x} P(X) = 1$

If the r.v is Cont
↓
Denoted by f(x) or f(X=x) and satisfies the following:

[1] $f(x) \geq 0$, $\forall x \in R_x$

[2] $\int_{R_x} f(x)\,dx = 1$

إذا ما تحقق هذان
الشرطان، & يكون الاقتران
(P.d.f)

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Prob. Ch. 2

* Main Rules.

p.d.f → r.v discrete

[1] $P(X \in A) = \sum_{x \in A} p(x)$, $A \subseteq R_X$

↳ $A = \{0, 1\} \rightarrow P(X \in A) = p(0) + p(1)$

[2] $P(X \in A \cap X \in B) = P(X \in A \cap B) = \sum_{x \in A \cap B} p(x)$.

↳ $A = \{0, 1, 2\}$ $B = \{1, 2, 3, 4\}$

$A \cap B = \{1, 2\}$

$P(X \in A \cap B) = p(1) + p(2)$

[3] $P(X \in A \mid X \in B) = \frac{P(X \in A \cap B)}{P(X \in B)} = \frac{\sum_{x \in A \cap B} P(x)}{\sum_{x \in B} P(x)}$

[4] The expected value of the r.v X is denoted by:

$E(X) = \sum_{x \in R_X} x P(x)$

[5] $E(g(x)) = \sum_{x \in R_X} g(x) \cdot p(x)$

↳ $E(x^2) = \sum_{x \in R_X} x^2 \cdot p(x)$

p.d.f → r.v cont

[1] $P(X \in A) = \int_A f(x)\, dx$

[2] $P(X \in A \cap X \in B) = P(X \in A \cap B) = \int_{A \cap B} f(x)\, dx$.

↳ $A = [0, 5]$ $B = [2, 7]$

$A \cap B = [2, 5]$

$P(X \in A \cap B) = \int_2^5 f(x)\, dx$.

[3] $P(X \in A \mid X \in B) = \frac{P(X \in A \cap B)}{P(X \in B)} = \frac{\int_{A \cap B} f(x)\, dx}{\int_B f(x)\, dx}$.

[4] The expected value of the r.v X is denoted by:

$E(X) = \int_{R_X} g(x) \cdot f(x)\, dx$.

[5] $E(g(x)) = \int_{R_X} g(x) \cdot f(x)\, dx$

↳ $E(x^2) = \int_{R_X} x^2 \cdot f(x) \cdot dx$.

prob. ch.2

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6 $E(aX+b) = aE(X)+b$

6 $E(aX+b) = aE(X)+b$.

7 The variance (التباين) of the r.v X is denoted by:

$Var(X) = E(X^2) - (E(X))^2$

7 The variance of the r.v X is denoted by:

$var(X) = E(X^2) - (E(X))^2$

* Note: variance → $Var(X)$, σX^2] → symbols of variance.

8 standard deviation.

$std(X) = \sqrt{Var(X)}$

$= \sigma X$

8 standard deviation

$std(X) = \sqrt{var(X)}$

$= \sigma X$.

9 $Var(aX+b) = a^2 Var(X)$

9 $Var(aX+b) = a^2 var(X)$.

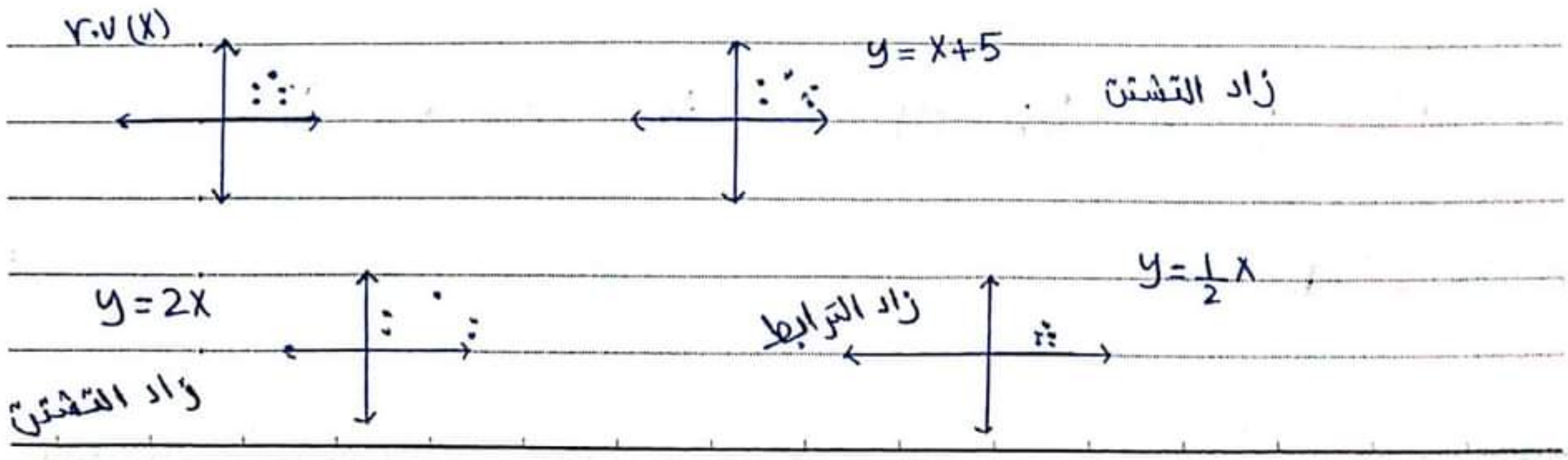
10 $std(aX+b) = |a| . std(X)$

10 $std(aX+b) = |a| . std(X)$.

Remark: $E(X) \equiv$ expectation of (X)

الوسط الحسابي $\equiv$ mean of (X)

$\equiv$ Average of (X) $= \bar{X}$



prob. CH.2 — Wed. 3. Nov. 2021

Remark: If X is a continuous r.v, then:

① $P(X=a) = \int_a^b f(x)\,dx = 0$

② $P(a \le X \le b) = \int_a^b f(x)\,dx$

Ex- Two balls are drawn from a box with 3 red and 2 black balls, Let X be the number of the red balls. Find: [1] E(X) [2] Var(X)

[3] $E(X^3)$

[4] If $y = -2X + 5$, Find: [a] E(X)

[b] Var(Y)

[c] Std(Y)

Sol: $R_X = \{0, 1, 2\}$ (X → discrete r.v)

[1] $E(X) = \sum_{X \in R_X} X\,P(X) = 0 \cdot P(0) + 1 \cdot P(1) + 2 \cdot P(2) = 1.2$

$P(0) = \frac{\binom{3}{0}\binom{2}{2}}{\binom{5}{2}} = \frac{1}{10} = 0.1$

$P(1) = \frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}} = \frac{6}{10} = 0.6$

$P(2) = \frac{\binom{3}{2}\binom{2}{0}}{\binom{5}{2}} = \frac{3}{10} = 0.3$

$0.1 + 0.6 + 0.3 = 1.2$

The prob. density function of r.v X

wed.

Prob. CH.2

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$$P(X=x) = \begin{cases} 0.1 & , X=0 \\ 0.6 & , X=1 \\ 0.3 & , X=2 \\ 0 & , \text{otherwise} \end{cases}$$

$P(x) \geq 0 \;/\; \sum_{x \in R_x} P(x) = 1$ ← يحقق الشرط

[2] $Var(X) = E(X^2) - (E(X))^2$

$= 1.8 - (1.2)^2$

$= \boxed{0.36}$

$E(X^2) = \sum_{x \in R_x} x^2 \cdot P(x) =$

$0 + 0.6 + 1.2 = 1.8$

[3] $E(X^3) = \sum_{x \in R_x} x^3 \cdot P(x) = 0 + 0.6 + 2.4 = \boxed{3}$

[4] (a) $E(Y) = E(-2X+5)$

$= -2\,E(X) + 5 = -2(1.2) + 5 = \boxed{2.6}$

(b) $Var(Y) = Var(aX+b)$

$= a^2 * Var(X)$

$= (-2)^2 * 0.36 = \boxed{1.44}$

(c) $Std(Y) = \sqrt{Var(Y)} = \sqrt{1.44} = \boxed{1.2}$

OR

$Std(Y) = Std(-2X+5)$

$= |-2| \cdot Std(X)$

$= 2 * \sqrt{Var(X)} = \boxed{1.2}$

Prob. Ch. 2

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Ex- Let X be a discrete r.v with $E(X) = 2$ and $Var(X) = 3$, Find $E(-2X^2 + 3X - 1)$

Sol: $Var(X) = E(X^2) - (E(X))^2$

$3 = E(X^2) - (2)^2 \Rightarrow E(X^2) = 7$.

$E(-2X^2 + 3X + 1) = -2E(X^2) + 3E(X) - 1$

$= -2 * 7 + 3 * 2 - 1 = \boxed{-9}$

Ex- Let X be a discrete r.v with p.d.f:

$$P(X) = \begin{cases} C, & X = 1, 2, 3, 4 \\ 0.1, & X = 5.6 \\ 0, & \text{otherwise} \end{cases}$$, Find the value of C?

Sol: $\sum_{x \in R_X} P(X) = 1$

$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$

$4C + 0.2 = 1$

$4C = 0.8 \Rightarrow \boxed{C = 0.2}$

Ex- Let X be a r.v with p.d.f:

$$P(X) = \begin{cases} C, & 0 \leq X < 1/2 \\ 2C, & 1/2 \leq X < 1 \\ 0, & \text{otherwise} \end{cases}$$, Find the value of C?

prob. CH.2

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Sol: $\int_{R_X} f(x)\,dx = 1$

$\int_0^{1/2} C.\,dx + \int_{1/2}^{1} 2C.\,dx = 1$

$1/2\,C + C = 1 \Rightarrow \boxed{C = 2/3}$

Ex- Let X be a cont rv with p.d.f:

$f(x) = C^2 . e^{-cx}$, $x > 0$ $(0 < X < \infty)$, Find:

1. The value of c if $c > 0$
2. $P(X \in (0,1))$
3. $P(X = 2)$
4. $E(X^2 + 2X - 1)$
5. $var(X)$
6. $E(2X - 3)$
7. $var(\sqrt{2}X + 12)$

Sol: 1. $\int_{R_X} f(x).\,dx = 1 \Rightarrow \int_0^{\infty} C^2 . e^{-cx}\,dx = 1$

$\Rightarrow \lim_{b\to\infty} \int_0^{b} C^2 . e^{-cx}\,dx = 1$

$\lim_{b\to\infty} \left[C^2 . \frac{1}{-C} e^{-cx} \right]_0^{b} = 1$

$-C \lim_{b\to\infty} (e^{-c.b} - 1) = 1$

$-C (e^{-c.\infty} - 1) = 1$

$-C (e^{-\infty} - 1) = 1 \Rightarrow -C(0 - 1) = 1$

$\boxed{C = 1}$ $f(x) = e^{-x}$.

2. $P(X \in (0,1)) = P(0 < X < 1)$

$\int_0^1 e^{-x}\,dx \Rightarrow -e^{-x}\Big]_0^1 = -e^{-1} + 1$

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prob. ch4.2

[3] $P(X=2) = \int_2^2 e^{-x}\,dx = 0$

[4] $E(X^2+2X-1)$

$E(X^2)$, $E(X)$ → "إيجادهم أولاً"

$$E(X) = \int_{Rx} X \cdot f(x)\,dx \Rightarrow \int_0^\infty X \cdot e^{-x}\,dx$$

⊕ X · e^{-x}
⊖ 1 · $-e^{-x}$
⊕ 0 · e^{-x}

$$\lim_{b\to\infty} \int_0^b X \cdot e^{-x}\,dx$$

$$\lim_{b\to\infty} \left[-Xe^{-x} + e^{-x}\right]_0^b$$

$$\lim_{b\to\infty} \left[-be^{-b} + e^{-b} - 1\right] \Rightarrow \lim_{b\to\infty} \frac{-b+1}{e^b} = \lim_{b\to\infty}(1)$$

$$= \lim_{b\to\infty} \frac{-1}{e^b} - 1 = \frac{-1}{e^\infty} - 1$$

$$= \frac{-1}{\infty} - 1 = 0 - 1 = \boxed{-1}$$

$$E(X^2) = \lim_{b\to\infty} \int_0^b x^2 \cdot e^{-x}\,dx$$

⊕ X^2 · e^{-x}
⊖ 2X · $-e^{-x}$
⊕ 2 · e^{-x}
⊖ 0 · $-e^{-x}$

$$\lim_{b\to\infty} \left[e^{-x}(-x^2 - 2x + 2)\right]_0^b$$

$$\lim_{b\to\infty} \left(\frac{-b^2 - 2b - 2 + 2}{e^b}\right)$$

$$\lim_{b\to\infty} \frac{-2b-2}{e^b} + \lim_{b\to\infty}(2)$$

$$\lim_{b\to\infty} \frac{-2}{e^b} + 2 = 0 + 2 = \boxed{2}$$

prob. ch.2

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$\therefore E(X^2 + 2X - 1) = E(X^2) + 2E(X) - 1$

$= 2 + 2(-1) - 1 = \boxed{-1}$

[5] $Var(X) = 2 - (-1)^2 = \boxed{1}$

[6] $E(2X - 3) = 2E(X) - 3$

$= 2(-1) - 3 = \boxed{-5}$

[7] $Var(\sqrt{2}X + 12) = (\sqrt{2})^2 \; Var(X)$

$= 2(1) = \boxed{2}$

Mon.
8. Nov. 2021

Def: Let X be a r.v with range R_X, then the Comulative Distribution function (C.d.f), or distribution function is given by:
توزيع التراكمي / التراكمي (C.d.f)

$$F(x) = \begin{cases} \sum_{u \le x} p(u) & \text{, If X is a discrete.} \\ \int_{-\infty}^{x} f(u) & \text{, If X is a cont.} \end{cases}$$

Remarks: ① F(x) is a step fun. when X is a discrete.

② F(x) is a cont. if X is cont.

③ $P(a \le X \le b) = F(b) - F(a)$ c.d.f:

If X is a cont:

p.d.f ④ $F(x) = \frac{d\,F(x)}{d(x)}$, If is a cont.

prob. CH.2

1ucn
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Ex: Let X is a discrete r.v with p.d.f is given by.

X	0	1	2
F(x)	1/4	1/2	1/4

. Find the following:

1. The c.d.f
2. $P(1 \leq X \leq 2)$
3. $P(0.5 < X < 1.5)$

Sol:

1. $$F(x) = \begin{cases} F(x) = 0 & , \quad X \leq 0 \\ F(x) = 0 + 1/4 = 1/4 & , \quad 0 \leq X < 1 \\ F(x) = 1/4 + 1/2 = 3/4 & , \quad 1 \leq X \leq 2 \\ F(x) = 3/4 + 1/4 = 1 & , \quad X \geq 2 \end{cases}$$

c.d.f is start of 0 and end the 1

$$F(x) = \begin{cases} 0 & , \quad X < 0 \\ 1/4 & , \quad 0 \leq X < 1 \\ 3/4 & , \quad 1 \leq X < 2 \\ 1 & , \quad X \geq 2 \end{cases}$$

2. $P(1 \leq X \leq 2) = P(1) + P(2) = 0.5 + 1/4 = \boxed{3/4}$

3. $P(0.5 < X < 1.5) = P(1) = 1/2$

له لو كان في مسألة، عزم يقرّها بالـ table

prob. CH. 2 — Mon. 8. Nov. 2021

Ex- Let X be a discrete r.v with c.d.f,

$$F(x) = \begin{cases} 0, & X<4 \\ 1/4, & 4\le X<5 \\ 1/2, & 5\le X<6 \\ 1, & X\ge 6 \end{cases}$$

Find The following:

1. The p.d.f:
2. $P(4\le X\le 6)$
3. $P(X=5.5)$

Sol.: 1) $R_X = \{4,5,6\}$

$P(X=4) = F(4) = 1/4$

$P(X=5) = F(5) - F(4) = 1/2 - 1/4 = 1/4$

$P(X=6) = F(6) - F(5) = 1 - 1/2 = 1/2$

$$P(X=x) = \begin{cases} 1/4, & x=4 \\ 1/4, & x=5 \\ 1/2, & x=6 \end{cases}$$

2) $P(4\le X\le 6) = P(4) + P(5) + P(6) = 1$

3) $P(X=5.5) = 0$ Not the range.

Ex. Let X be a cont r.v with p.d.f $f(x) = e^{-\lambda x}$,

$X\in(0,\infty)$. Find: 1) The c.d.f

$X>0$

2) $P(1\le X\le 2)$

3) $P(1<X<2)$.

Sol.: 1)

$$F(x) = \int_{-\infty}^{x} f(u)\,du = \int_{0}^{x} e^{-\lambda u}\,du = \frac{1}{-\lambda} e^{-\lambda u}\Big]_0^x$$

$$= \frac{-1}{\lambda}\left(e^{-\lambda x} - 1\right) \quad \text{c.d.f}$$

2) $P(1\le X\le 2) = \int_1^2 f(x)\,d(x)$

OR

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$= F(2) - F(1) = \frac{-1}{\lambda}\left[e^{-2\lambda} - 1\right] + \frac{1}{\lambda}\left[e^{-\lambda} - 1\right]$

[3] $P(1<X<2)$ → same the ②

[4] $P(X=2) = 0$ Cont $\int_2^2 F(x)\,dx = 0$.

Ex: Let X be a cont r.v with c.d.f

$F(x) = 1 - \ln(1-\alpha x)$. Find:

[1] The p.d.f [2] $P(0<X<0.5)$

Sol: [1] $f(x) = \frac{dF(x)}{dx} = 0 - \frac{0-\alpha}{1-\alpha x} = \frac{\alpha}{1-\alpha x}$

[2] $P(0<X<0.5) = \int_0^{0.5} \frac{\alpha}{1-\alpha x}\,dx = \alpha \frac{1}{-\alpha} \ln(1-\alpha x)\Big|_0^{0.5}$

$= -\ln(1-0.5x)$

$= -\ln(1-0.5\alpha)$.

or $P(0<X<0.5) = F(0.5) - F(0)$

$= 1 - \ln(1-0.5\alpha) - (1 - \ln(1))$

$= -\ln(1-0.5\alpha)$.

CH.3 : Bivariate Random variables

متغيران العشوائية الثنائية.

Def: Let X and Y be two Random variables with Ranges R_X and R_Y, then the two Dimentional Range of X and Y is given by

$$R_{XY} = \{ (x,y) \mid x \in R_X , y \in R_Y \}$$, with

joint p.d.f , $f(x,y) = P\{X=x , Y=y\}$, which satisfies the following: ∩ intersection

1) $f(x,y) \geq 0$, $\forall (x,y) \in R_{XY}$.

2) $\sum_{x \in R_X} \sum_{y \in R_Y} f(x,y) = 1$, $\iint_{R_{XY}} f(x,y)\, dx\, dy = 1$.

discreted	Cont
If the r.v is discrete	If the r.v is cont.
2) $\sum_{x \in R_X} \sum_{y \in R_Y} f(x,y) = 1$	2) $\iint_{R_{XY}} f(x,y)\, dx\, dy = 1$
3) The marginal p.d.f X is given by: $f_X(x) = \sum_{y \in R_Y} f(x,y)$	3) The marginal p.d.f X is given by: $f_X(x) = \int_{R_Y} f(x,y)\, dy$. تكامل بدلالة y
4) The marginal p.d.f y is given by: $f_Y(y) = \sum_{x \in R_X} f(x,y)$ تكامل بدلالة x ، اقتران الكثافة العشوائية للمتغير y	4) The marginal p.d.f y is given by: $f_Y(y) = \int_{R_X} f(x,y)\, dx$ "
5) $E(X) = \sum_{x \in R_X} x\, f_X(x)$	5) $E(X) = \int_{R_X} x\, f_X(x)\, dx$.
6) $E(Y) = \sum_{y \in R_Y} y\, f_Y(y)$	6) $E(Y) = \int_{R_Y} y\, f_Y(y)\, dy$.

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[7] $E(X,Y) = \sum_{x \in R_X} \sum_{y \in R_Y} xy\, f(x,y)$

OR: $E(g(x,y)) =$

$\sum_{x \in R_X} \sum_{y \in R_Y} g(x,y)\, f(x,y)$

[7] حالة خاصة من

[8] The covariance of x and y ps. given by:

$Cov(X,Y) = E(XY) - E(X)\,E(Y)$

تباين المشترك

[9] $Cov(aX+b, cY+d) =$

$ac\ Cov(X,Y)$

[10] The correlation coeffecient at X and Y Ps given by

$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_X\, \sigma_Y}$

$\sigma_X = Std(X)$

$\sigma_Y = Std(Y)$

[7] $E(X,Y) = \iint_{R_{XY}} xy\, f(x,y)\, dx\, dy$

OR: $E(g(x,y)) =$

$\iint_{R_{XY}} g(x,y)\, f(x,y)\, dy\, dx.$

[8] نفس القانون بلا

⇐ cont, discrete

[9] $Cov(aX+b, cY+d) =$
$ac\ Cov(X,Y).$

[10]

[8] ⇒ Same.

$Cov.(aX+b, cY+d) = E[(aX+b)\cdot(cY+d)] -$

$E(aX+b) - E(cY+d).$

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Remarks :

① $-1 \leq Corr(X,Y) \leq 1$

② $Corr(aX+b, cY+d) =$ $\pm$ $\pm$

$\begin{cases} Corr(X,Y) & \text{, if } a,c \text{ have the same sign.} \\ -Corr(X,Y) & \text{, if } a,c \text{ have the different sign.} \end{cases}$ $\pm$ $\pm$

Remarks : If X and Y are independent, then:

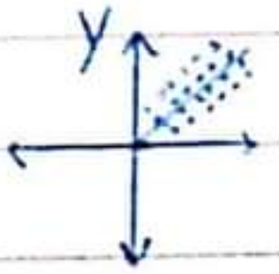
① $f(x,y) = P(X=x, Y=y)$.
$= P(X=x) \cdot P(Y=y)$.
$= f_X(x) \cdot f_Y(y)$

② $E(X,Y) = E(X) \cdot E(Y)$

③ $Cov(X,Y) = E(XY) - E(X) \cdot E(Y)$
$= E(X) \cdot E(Y) - E(X) \cdot E(Y) = 0$

④ $Corr(X,Y) = 0 \Rightarrow \frac{Cov(X,Y)}{\sigma_X \cdot \sigma_Y} = \frac{0}{1} = 0$

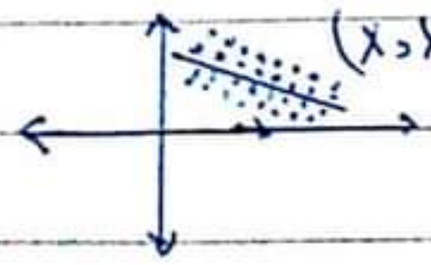
Remark : ① If $Corr(X,Y) \simeq 1$, then:



Strong relation between X and Y

العلاقة طردية. $\Rightarrow$ تزيد القيم

② If $Corr(X,Y) \simeq -1$, then:



$\Rightarrow$ Strong relation between X and Y

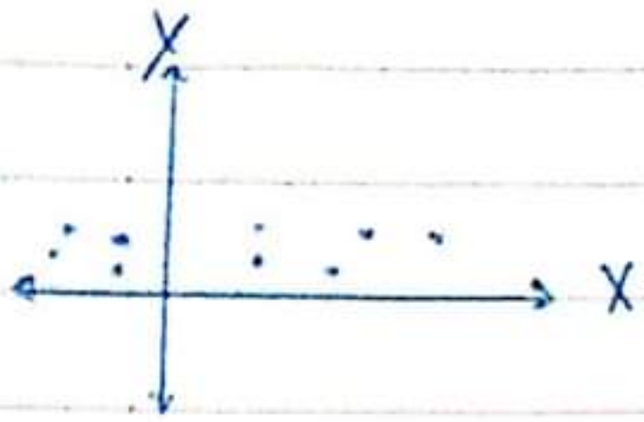
العلاقة عكسية.

تقل القيم

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③ If $Corr(X,Y) = 0$, then



No relation.

Independant ④ $Corr(X,Y) = 0$

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EX- Three of five items are defective such that one item defective in size and two items are defective in design. Two items are drawn randomly.

X: number of defective in size items.

Y: number of defective in design items.

Find the following: ① $E(3+4X)$.

② $E(X^2+3X-5)$

③ $var(2-4X)$. ④ $var(Y)$.

⑤ $E(XY)$ ⑥ Are X and Y independent?.

Sol: $X = \{0,1\}$, $Y = \{0,1,2\}$

$R_{XY} = \{(0,0), (0,1), (0,2),$
$(1,0), (1,1), (1,2)\}$.

$n(S) = \binom{5}{2} = \frac{5!}{2!\,3!} = \boxed{10}$

$f(0,0) = \frac{\binom{1}{0}\binom{2}{0}\binom{2}{2}}{\binom{5}{2}} = 0.1$

prob. CH.3

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$$f(0,1) = \frac{(1)(2)(2)}{10} = 0.4$$

$$f(0,2) = \frac{(1)(2)(1)}{10} = 0.1$$

order is not important.

$$f(1,0) = \frac{(1)(2)(2)}{10} = 0.2$$

$$f(1,1) = \frac{(1)(2)(2)}{10} = 0.2$$

$$f(1,2) = 0 \quad \text{impossible}$$

عدد الطلاب بالسؤال 2

$$E(X) = \sum_{x \in R_X} x f_X(x).$$

y \ X	0	1
0	0.1	0.2
1	0.4	0.2
2	0.1	0

joint P.d.f

$$f_X(x) = \sum_{y \in R_Y} f(x,y)$$

$$= f(x,0) + f(x,1) + f(x,2)$$

$$f_X(0) = f(0,0) + f(0,1) + f(0,2) = \boxed{0.6}$$

$$f_X(1) = 0.4$$

The marginal p.d.f of (X).

X	$f_X(x)$	$x f_X(x)$	$x^2 f_X(x)$
0	0.6	0	0
1	0.4	0.4	0.4
		0.4	.0.4

1) $E(3+4X) = 3 + 4E(X) = 3 + 4(0.4) = \boxed{4.6}$

2) $E(X^2 + 3X - 5) = E(X^2) + 3E(X) - 5$

$= 0.4 + 3(0.4) - 5 = \boxed{-3.4}$

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[3] $\text{var}(2-4X) = (-4)^2 \text{ var}(X)$

$= 16\,(E(X^2) - (E(X))^2$

$= 16\,(0.4 - 0.16) = 3.84$

[4] $f_{Y(y)} = \sum_{x \in R_X} f(x,y) = f(0,y) - f(1,y)$.

y	$f_Y(y)$	$y.f_Y(y)$	$y^2 f_Y(y)$
0	0.3	0	0
1	0.6	0.6	0.6
2	0.1	0.2	0.4
	1	0.8	1

$f_Y(0) = f(0,0) + f(1,0)$

$= 0.3$

$\text{var}(Y) = 1 - (0.8)^2 = 1 - 0.64 = 0.36$

(X,Y)	XY	f(X,Y)	XY f(X,Y)
(0,0)	0	f(0,0) = 0.1	0
(0,1)	0	f(0,1) = 0.4	0
(0,2)	0	0.1	0
(1,0)	0	0.2	0
(1,)	1	0.2	0.2
(1,2)	2	0	0

[5] $F(X,Y) = \sum_{x \in R_X} \sum_{y \in R_Y} XY\, f(x,y) = 0.2$

[6] $E(X) = 0.4$, $E(Y) = 0.8$ not indep.

$E(XY) \stackrel{?}{=} E(X) \cdot E(Y)$

$0.2 \neq 0.32$. X, Y dep.

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prob. CH.3

□ Cov(2X-3 , 3-4Y)

$= (2)(-4)\ Cov(X,Y)$

$= -8\ (E(XY) - E(X).E(Y))$

$= -8\ (0.2 - 0.32) = \boxed{0.96}$

□ Corr.(2X-3 , 3-2Y)

$$-1 \cdot \frac{Cov(X,Y)}{\sqrt{Var(X)} \cdot \sqrt{Var(X)}} = -1 \cdot \frac{0.12}{\sqrt{0.24} \cdot \sqrt{0.36}}$$

* Def: The joint comulative distribution function [joint c.d.f] is given by:

$F_{XY}(x,y) = P\{X \leq x , Y \leq y\}$. and can be calculated as:

$$F_{XY}(x,y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f(u,v)\,du\,dv$$

properties:

1. $0 \leq F_{XY}(x,y) \leq 1$ (عند أدنى قيمة ... عند أعلى قيمة)
2. $f(x,y) = \frac{\partial^2 F(x,y)}{\partial x\, dy}$
3. $F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$.
4. $F_X(y) = \int_{-\infty}^{y} f_Y(v)\,dv$.
5. $f_X(x) = \frac{dF_X(x)}{dx}$
6. $f_Y(y) = \frac{dF_Y(y)}{dy}$

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1. $P(X \in A, Y \in B) = \iint_{B\ A} f(x,y)\, dx\, dy$

OR: $P(x_1 \le X \le x_2, y_1 \le Y \le y_2)$

$A_1 = [x_1, x_2]$

$$= \int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x,y)\, dx\, dy$$

2. $\iint_{Rxy} f(x,y)\, dx\, dy = 1$

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Ex.- Let $f(x,y) = \begin{cases} b e^{-x} \cos y, & 0 \le x \le 2,\ 0 \le y \le \pi/2 \\ 0, & \text{otherwise.} \end{cases}$

Find the following:

1. The value of b.
2. The joint c.d.f $F_{XY}(x,y)$
3. $F_X(x)$
4. $F_Y(y)$
5. $P\{0 \le X \le 1,\ 0 \le y \le \pi/4\}$.

Sol: ① $\iint_{Rxy} f(x,y)\, dx\, dy = 1$

$$\int_0^{\pi/2} \left[\int_0^2 b e^{-x} \cos y\, dx \right] dy = 1$$

بالنسبة لـ x ثابت

$$\int_0^{\pi/2} b \cdot \cos y \left[-e^{-x} \right]_0^2 dy = 1$$

prob. CH.3

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$$\int_0^{\pi/2} b.(-e^{-2}+1).\cos y\,dy = 1$$

$$b(-e^{-2}+1)\sin y\Big]_0^{\pi/2} = 1$$

$$b(-e^{-2}+1) = 1 \Rightarrow b = \frac{1}{1-e^{-2}}$$

$$f(x,y) = \frac{1}{1-e^{-2}} \cdot e^{-x} \cdot \cos y \quad , \quad 0 \le x \le 2 \ , \ 0 \le y \le \pi/2.$$

[2] $$F_{XY}(x,y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f(u,v)\,du\,dv$$

$$= F_{XY}(x,y) = \int_0^y \left[\int_0^x \frac{1}{1-e^{-2}} \cdot e^{-u} \cdot \underbrace{\cos v}_{\text{ثابت}}\,du\right]dv$$

$$= \int_0^y \frac{1}{1-e^{-2}} \cdot \cos v \left[-e^{-u}\right]_0^x \cdot dv$$

$$= \int_0^y \frac{1}{1-e^{-2}} \cdot (-e^{x}+1) \cdot \cos v\,dv$$

$$= \frac{1}{1-e^{-2}} \cdot (-e^{-x}+1) \cdot \sin v\Big]_0^y$$

$$= \frac{(1-e^{-x})\cdot \sin y}{1-e^{-2}}$$

$$x$$

[3] $$F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$$

$$f_X(x) \int_{R_Y} f(x,y)\,dy = \int_0^{\pi/2} \frac{1}{1-e^{-2}} \cdot e^{-x} \cos y\,dy$$

$$= \frac{e^{-x}}{1-e^{-2}} \sin y\Big]_0^{\pi/2} = \frac{e^{-x}}{1-e^{-2}}$$

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$$F_X(x) = \int_0^x \frac{e^{-u}}{1-e^{-2}}\,du = \frac{1}{1-e^{-2}}\,-e^{-u}\Big]_0^x$$

$$= \frac{1}{1-e^{-2}}\cdot(e^{-x}+1)$$

[4] $$F_Y(y) = \int_0^2 \frac{1}{1-e^{-2}}\cdot e^{-x}\cos y\cdot dx$$

$$= \frac{1}{1-e^{-2}}\cos y\ -e^{-x}\Big]_0^2 = \frac{\cos y}{1-e^{-2}}\,(-e^{-2}+1).$$

$$F_Y(y) = \int_0^y \cos v\,dv = \sin v\Big]_0^y = \sin y$$

Ex. Evaluate:

① $$\int_0^1\int_0^2 xy^2\,dx\,dy$$

$$= \int_0^1 y^2\ \frac{x^2}{2}\Big]_0^2\,dy$$

$$= \int_0^1 y^2\ (2-0)\,dy = \int_0^1 y^2\,2\,dy = \frac{2y^3}{3} = \boxed{2/3}$$

② $$\int_0^2\int_0^1 xy^2\,dy\,dx$$

$$= \int_0^2 x\frac{y^3}{3}\Big]_0^1\,dx = \int_0^2 x\,(1/3-0)\,dx$$

$$= 1/3\cdot\frac{x^2}{2}\Big]_0^2 = 1/3\,(2-0) = 2/3$$

Rule: $$\int_c^d\int_a^b f(x,y)\,dx\,dy = \int_a^b\int_c^d f(x,y)\,dy\,dx.$$

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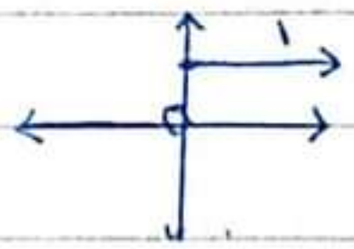
unit step function :

Def : The unit step fun is defined by:

① $u(x-c) = \begin{cases} 1 & , x \geq c \\ 0 & , x < c \end{cases}$

properties :

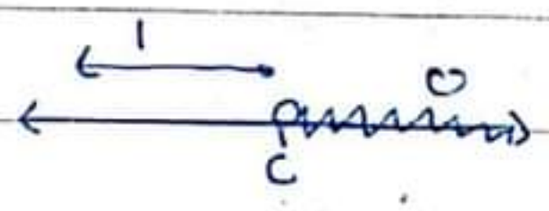
① $u(x) = \begin{cases} 1 & , x \geq 0 \\ 0 & , x < 0 \end{cases}$



② $f(x) \cdot u(x) = \begin{cases} f(x) & , x \geq 0 \\ 0 & , x < 0 \end{cases}$

③ $f(x) \cdot u(x-c) = \begin{cases} f(x) & , x \geq c \\ 0 & , x < c \end{cases}$

④ $u(c-x) = \begin{cases} 0 & , x > c \\ 1 & , x \leq c \end{cases}$



Def: The unit impulse fun. is denoted by $\delta(x)$ and satisfied the following:

① $\int_{-\infty}^{\infty} \phi(x)\,\delta(x-c)\,dx = \phi(c)$

$\phi(x) = 1$

$\phi(c) = 1$

② $\int_{-\infty}^{\infty} \delta(x-c)\,dx = 1$

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③ $\frac{du(x)}{dx} = \delta(x)$.

④ Let $X \in \{x_1, x_2 \dots x_n\}$ be the discrete the r.v, then :

[a] The prob. density function ps :

$$p(x) = p(x_1)\delta(x-x_1) + p(x_2)\delta(x-x_2) \dots + p(x_n)\delta(x-x_n).$$

[b] ps :

$$F(x) = p(x_1)u(x-x_1) + p(x_2)u(x-x_2) \dots + p(x_n)u(x-x_n).$$

Ex- Evaluate : ① $\int_{-\infty}^{u} \underbrace{x^2}_{\phi(c)} \cdot \underbrace{\delta(x-2)}_{c} dx$

$= (2)^2 = 4$

② $\int_{-\infty}^{\infty} \delta(x)\,dx = 1 \Rightarrow \int_{-\infty}^{\infty} (x+2)\delta(x)\,dx = 0+2 = \boxed{2}$

Ex- Let x be a discrete r.v with $R_x = \{-1, 1\}$
$p(-1, 1/2)$ and $p(1) = 1/2$. Find the following :

[1] The p.d.f [2] The c.d.f

$x_1 = -1$
$x_2 = 1$

Sol.: [1] $p(x) = p(-1).\delta(x+1) + p(1)\delta(x-1)$

$= 1/2\,\delta(x+1) + 1/2\,\delta(x-1)$.

[2] $1/2\,u(x+1) + 1/2\,u(x-1)$.

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Ex - If the joint p.d.f is given by:
$f(x,y) = u(x)\,u(y)\,xe^{-x(y+1)}$, Find the following:
① $F_X(x)$ ② $F_Y(y)$ ③ $F_{XY}(x,y)$.

Sol: $u(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$

$u(y) = \begin{cases} 1, & y \geq 0 \\ 0, & y < 0 \end{cases}$

Cont r.v
Bivariate.

① $f_X(x) = \int_{Ry} f(x,y)\,dy = \int_{-\infty}^{\infty} f(x,y)\,dy =$

$$\int_{-\infty}^{\infty} u(x)u(y)\,xe^{-x(y+1)}\,dy = \left[\int_{-\infty}^{0} u(x)\cdot 0\cdot xe^{-x(y+1)}\,dy\right] + \lim_{b\to\infty}\int_{0}^{b} u(x)\cdot 1\cdot xe^{-x(y+1)}\,dy$$

(Zero: $y<0$; $y\geq 0$)

$$= \lim_{b\to\infty} u(x)\cdot x\cdot\frac{1}{-x}\,e^{-x(y+1)}\Big|_0^b$$

$$= -\lim_{b\to\infty} u(x)\left[e^{-x(b+1)} - e^{-x(0+1)}\right]$$

$$= -u(x)\left[e^{-x(\infty)} - e^{-x}\right] \qquad \boxed{e^{-\infty} = 0}$$

$$= -u(x)\,e^{-x} = f_X(x)$$

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$$Fx(x) = \int_0^x f_X(u)\,du$$

$$= \int_0^x u(z)\,e^{-z}\,dz$$

$$\int_{-\infty}^{x} = \int_{-\infty}^{0} + \int_0^x \quad (0 \text{ is the zero})$$

$$= \int_0^x 1 \cdot e^{-z}\,dz = -e^{-z}\Big]_0^x = \boxed{-e^{-x}+1}$$

[2] Marginal p.d.f of y

$$F_Y(y) = \int_{R_X} f(x,y)\,dx$$

$$= 0 + \lim_{b\to\infty} \int_0^b 1 \cdot u(y)\, x e^{-x(y+1)}\,dx.$$

$$= \lim_{b\to\infty} u(y) \int_0^b x e^{-x(y+1)}\,dx \quad \rightarrow$$

$$= \lim_{b\to\infty} u(y) \cdot \frac{-x}{y+1} e^{-x(y+1)} - \frac{1}{(y+1)^2} e^{-x(y+1)} \Big]_0^b$$

$$\oplus x \quad -e^{-x(y+1)}$$

$$\ominus 1 \quad \frac{-1}{y+1} e^{-x(y+1)}$$

$$\oplus 0 \quad \frac{1}{(y+1)^2} e^{-x(y+1)}.$$

$$= \lim_{b\to\infty} u(y) \left(\frac{-b}{(y+1)\cdot e^{b(y+1)}} - \frac{e^{-b(y+1)}}{(y+1)^2} \right) - \left(0 - \frac{1}{(y+1)^2} \right)$$

$$= \left(0 - 0 + \frac{1}{(y+1)^2} \right) u(y) = \frac{u(y)}{(y+1)^2}$$

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Marginal C.d.f :

$$F_Y(y) \int_0^y f_Y(v)\,dv = \int_0^y \frac{u(v)}{(v+1)^2}\,dv$$

$$= \int_0^y (v+1)^{-2}\,dv = \frac{(v+1)^{-1}}{-1}\Big]_0^y = \frac{-1}{v+1}\Big]_0^y$$

$$= \frac{-1}{y+1} + 1$$

[3] $F_{XY}(x,y) = \iint_{RX} f_{-}(z,v)\,dz\,dv$ $\quad y=v,\ x=z$

$$= 0 + \int_0^y \boxed{\int_0^x \underbrace{u(z)}_{\text{II}} \cdot u(v) \cdot ze^{-z(v+1)}\,dz}\,dv$$

$$= \int_0^y \int_0^x u(v) \cdot \textcircled{z} e^{-z(v+1)}\,dz\,dv$$

$$= \int_0^y u(v) \left[\frac{-z}{v+1} \cdot e^{-z(v+1)} - \frac{1}{(v+1)^2} e^{-z(v+1)}\right]_0^x dv$$

$$= \int_0^y u(v) \left(\frac{-x}{v+1} e^{-x(v+1)} - \frac{1}{(v+1)^2} e^{-x(v+1)}\right) - \left(0 - \frac{1}{(v+1)^2}\right) dv$$

$\oplus z \cdot e^{-z(v+1)}$

$\ominus 1 \rightarrow \frac{-1}{v+1} e^{-z(v+1)}$

$\oplus 0 \rightarrow \frac{1}{(v+1)^2} e^{-z(v+1)}$

هاد الفرع هيك ممكن يجيبه الدكتور بدون z

امتحان امين - لهون

prob. CH.4 — Mon. 22. NOV. 2021

CH.4 - probability Distribution.
التوزيعات الاحتمالية.

Sec.4.1 - Binomial Dist.
التوزيع ذو حدين

- **Binomial Dist :**

It is a dist. of a randomized experiment with only two outcomes, one of which is the success of the experiment, and the other of it's failure.
توزيع لتجربة عشوائية، لها نتيجتان فقط أحدهما نجاح التجربة، والآخر فشلها.

- n.: The number of indep. and identical trails.
عدد المحاولات المستقلة المتماثلة.
- X : The number of success out of n trails, such that :
n: عدد المحاولات
$Rx = \{0, 1, 2, 3, \dots, n\}$ X = عدد النجاح
- $p = P(\text{success})$ احتمال النجاح في أي محاولة.
- $q = P(\text{fail}) = 1 - p$ احتمال الفشل في أي محاولة.

Def: The binomial r.v X is denoted by :
two parameters.

$X \sim B(n, p)$ with p.d.f
(X تتوزع) — Binomial

$$P(X = x) = \binom{n}{x} p^x q^{(n-x)}, \quad x = 0, 1, \dots, n.$$

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Such that : ① $E(X) = np$

② $var(X) = npq$

Note that : Binomial r.v is Discrete.

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wed

Ex- In a multiple choice exam with 10 questions that each question has 5 answers with only one true answer.

① Find the prob. that you will answer only 5 true.

② Find the prob. that you will get a fail mark.

③ Find the prob. that you will answer at most 2 Q. Correctly.

④ Find the mean of true answered questions.

⑤ Find the variance of true answered questions.

Sol : X : The number of true answered questions.

$R_X = \{0, 1, 2, \dots 10\} \Rightarrow n = 10$

$p = \frac{1}{5} = 0.2$ $\quad q = 1 - 0.2 = 0.8$

$X \sim B(10, 0.2)$ — ممكن يجيب هاد السؤال وعليه كل أسئلة بعد اللي فوق

$P(X = x) = \binom{10}{x} (0.2)^x (0.8)^{10-x}$

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Such that : 1) $E(X) = np$

2) $var(X) = npq$

Note that : Binomial r.v is Discrete.

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ice exam with ·o
stion has 5 answers with

u will answer only 5 true.
u will get a fall mark.
n will answer at most 2 Q.
Correctly.

(4) Find the mean of true answered questions.
(5) Find the variance of true answered questions.

Sol : X : The number of true answered questions.

$R_X = \{0, 1, 2, \dots 10\} \Rightarrow n = 10$

$p = \frac{1}{5} = 0.2$ $\quad q = 1 - 0.2 = 0.8$

$X \sim B(10, 0.2)$ — ممكن يجيب هاد السؤال وعليه 5 أسئلة بدل اللي فوق

$P(X = x) = \binom{10}{x} (0.2)^x (0.8)^{10-x}$

١٠ أسئلة
كل سؤال 5 خيارات
١) كل سؤال بتجاوبه بشكل مستقل indep.
احتمال خاطئ $p = 1/5 = 0.2$
٢) Identical ← احتمال النجاح في كل محاولة متساوي

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① $P(X=5) = \binom{10}{5}(0.2)^5(0.8)^{5}$ ~ 10-5

② $P(X=10) = \binom{10}{10}(0.2)^5(0.8)^0 = (0.2)^{10}$

③ $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$
بعوّضنا عادي حتى p.d.f.

④ $E(X) = np = 10*(0.2) = 2$

⑤ $V(X) = npq = 10(0.2)(0.8) = 1.6$

⑥ $E(2X^2) = 2E(X^2)$ ⇒ $Var(X) = E(X^2) - (E(X))^2$

$1.6 = E(X^2) - (2)^2$

$E(X^2) = 5.6$.

→ $Std(X) = \sqrt{1.6}$

Sec. 4.2 - Poisson Dist توزيع بوسون

عدد التكرارات لكل وحدة زمن

X : The number of accurence per unit of time.

m : average of accurence per unit of time.

Ex.- X : The number of patients entering the ER per hour.

$X = \{0, 1, 2, \ldots\}$ عدد صحيح Range

Discrete [poisson

Average الوسط

$0 \to \infty$

$[0 \to n]$ ← Binomial الوسط

n p

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Def: If $X \sim P(m)$, then the [p.d.f] of poisson dist. is given by:

$$P(X=x) = \frac{e^{-m} \cdot m^x}{x!}, \quad x = 0, 1, 2, \ldots \text{such that}$$

m: معدّل التكرار — الوسط الحسابي

$$\boxed{E(X) = Var(X) = m}$$
$$E(x^2) = m + m^2$$

EX-. Assume that the average number of patients entering emergency room is 1.2 per hour,

① Find the prob. that only 2 patients will enter per hour.
② Find the prob. that at least 2 patients will enter per hour.
③ Find the prob. that the number of patients is between 2 and 3 per hour.
④ Find $P(1.1 < X < 2.9)$
⑤ $P(X = 2.5)$
⑥ $E(2X+1)$
⑦ $std(-3X+4)$.

Sol: $m = 1.2 \Rightarrow X \sim P(1.2) \Rightarrow$

$$P(X=x) = \frac{e^{-1.2} \cdot (1.2)^x}{x!} \Rightarrow \text{p.d.f}, \quad x = 0, 1, 2, \ldots$$

① $P(X=2) = \frac{e^{-1.2} \cdot (1.2)^2}{2!}$

② $P(X \geq 2) = P(X=2) + P(X=3) + \ldots$ مش معطيني حد، مش معطيني حد

③ $P(2 \leq X \leq 3) = P(X=2) + P(X=3)$

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(4) $P(1.1 < X < 2.9) = P(X = 2)$

(5) $P(X = 2.5) = Zero$ (impossible) $2.5 \notin R_X = \{0, 1, 2, 3, \dots\}$.

(6) $E(2X+1) = 2E(X) + 1 = 2(1.2) + 1 = \boxed{3.4}$

$\because m = 1.2 = E(X)$

(7) $Std(-3X+4)$

$\because Var(X) = 1.2 \Rightarrow Var(-3X+4)$

$\because = (-3)^2 Var(X) = 9(1.2)$

$= 1.08$

$Std(-3X+4) = \sqrt{1.08}$

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CH.4

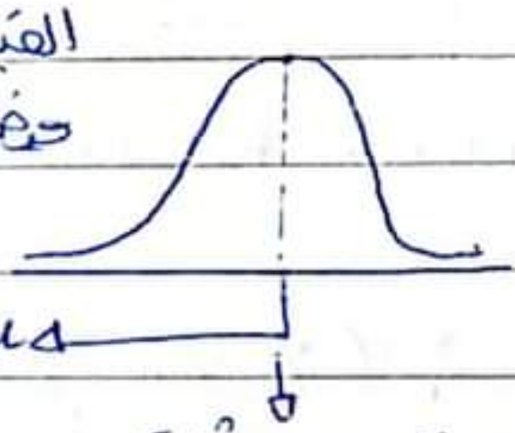
Normal Distribution التوزيع الطبيعي
(Gaussian dist).
(Bell-shaped dist)

يعتمد على
- mean: الوسط الحسابي
- التباين: σ^2

- Normal random variable is cont

properties :-

1) mean = E(X) = median = mode
القيمة التي تتساوى فيها الأعلى مع الأدنى
له الثوال: القيمة الأكثر تكرارًا
denoted by μ
القيم متماثلة على اليمين وعلى اليسار.



2) symmetric dist. نقطة التماثل.

3) The normal r.v X depends only on μ and σ^2, so we denoted it by:

$$X \sim N(\mu, \sigma^2)$$
normal — mean — var

4) If $X \sim N(\mu, \sigma^2)$, then:

a) $E(X) = \mu$, $Var(X) = \sigma^2$ Cont.

p.d.f

b) $f(X) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$
Lo std

$-\infty < X < \infty$

لا يمكن لا σ أو Var(X) يكون سالب

c) $P(a \leq X \leq b) = P(a < X < b)$ cont التكامل ما يغير

$= \int_a^b f(x)\,dx$ = area under $f(x)$ between $(x=a)$ and $(x=b)$.

p.d.f تكامل

$= F(b) - F(a) \Rightarrow$ F(X) تابع التكامل

$= \int_{-\infty}^{b} f(x)\,dx - \int_{-\infty}^{a} f(x)\,dx$

c.d.f هو هي المساحة على يسار العدد

= area to the left of b - area to the left of a.

d) $P(X = C) = 0$ [التكامل عند أي نقطة بساوي 0 $\int_c^c$]

e) $P(X \leq b) = \int_{-\infty}^{b} f(x)\,dx = F(b)$ = area to the left of b.

* Standard Normal Dist

حالة خاصة من normal dist

Let $X \sim N(\mu, \sigma^2)$, then the r.v:

$Z = \frac{X - \mu}{\sigma} \sim N(0,1)$ = std Normal Dist

$\sigma \rightarrow$ std

$\rightarrow$ new r.v

التوزيع الطبيعي المعياري

$\mu = 0$

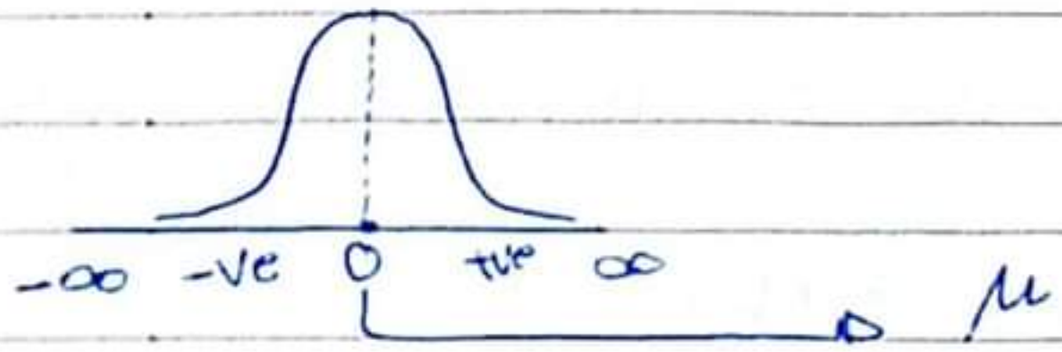
$\sigma^2 = 1$

تم إيجاده بالاعتماد على المتغير العشوائي الأصلي.

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properties :



من الاقتران الاحتمالي $f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{\frac{1}{2\sigma^2}(x-\mu)^2}$

وبهوفا ومانسكا

$Var(X) = 1$

$\sigma^2 = 1$

$-\infty < Z < \infty$

[2] symmetric about 0.

[3] $E(Z) = 0$, $Var(Z) = 1$

[4] $f(Z) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}Z^2}$

[5] $P(a \le Z \le b) = \int_a^b f(Z)\, dZ$ = area under $f(Z)$ between a and b

dist is cont
نجده باستخدام التكامل

$= T(b) - T(a)$

table ← table at b − table at a.

= area to the left b − area to the left a.

F شكل عام C.d.f
إذا بار std بتكون
T

[6] $\int_{-\infty}^{\infty} f(Z)\, dZ = 1$ total area.

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7) $\int_{-\infty}^{0} f(z)\,dz = 0.5 = \int_{0}^{\infty} f(z)\,dz$

لأنه متماثل حول النقطة صفر

8) $P(Z \leq b) = T(b)$

لازم يكون جاهزة في table ← لو بدون table

with p.df $\int_{-\infty}^{b} f(z)\,dz$.

9) $P(Z \geq b) = 1 - P(Z < b) = 1 - T(b)$.

with p.d.f $\Rightarrow \int_{b}^{\infty} f(z)\,dz$.

صعب تكامله

لأنه ما الله قيمة صاح تمامًا بتكون صعبة تقريبية

كيف نجده؟

باستخدام الجدول المعطى كما نتائج تقريبية للتكاملات التي حدودها

b
$\int$
$-\infty$

10) $P(Z = b) = 0$

الاحتمال عند نقطة حتى Zero دائمًا cont r.v

prob. CH.4. Mon 6. Dec. 2021

Ex)① Find $p(Z<1.15) = T(1.15)$

$= 0.87493$ ← من الجدول

[يمثل العدد بالجدول الرقم الصحيح + الجزء العشري] ← 1.15 ← 0.05 ← الجزء بالمئة يمثل الصف

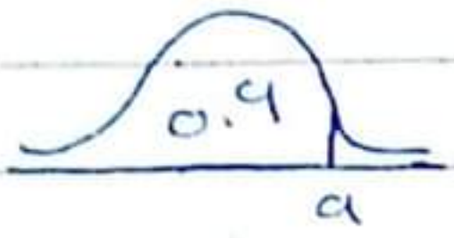
Ex)② $p(-1.25 < Z < 2.15)$ ← 5 عكس الكاتبة ← $T(b) - T(a)$

$T(2.15) - T(-1.25)$

$= 0.98422 - 0.10565 = 0.87857$

Ex)③ Find a s.t ← such that $p(Z<a) = 0.9$

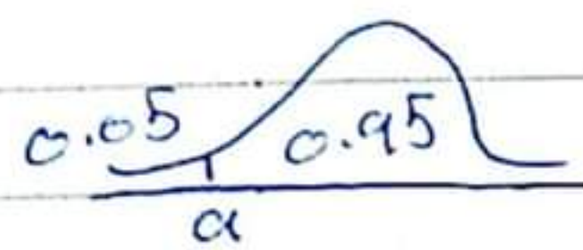
← يمر عبر قيمها في الجدول (أو)



بين الأرقام ← $a = 1.28$

ويختار أقرب الذي لها.

Ex)④ Find a s.t $p(Z>a) = 0.95$



$1 - p(Z \le a) = 0.95$

$-1 \quad -1$

$p(Z \le a) = 0.05$

← يمر عبر هوا الجدول على 0.05

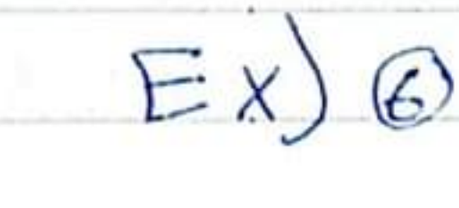
$a = -1.64$

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Ex) ⑤ Find a s.t

$P(-a < z < a) = 0.95$ ← بطرحها من 1 وتقسم على 2 ← 0.025 بتطلع



we want a s.t

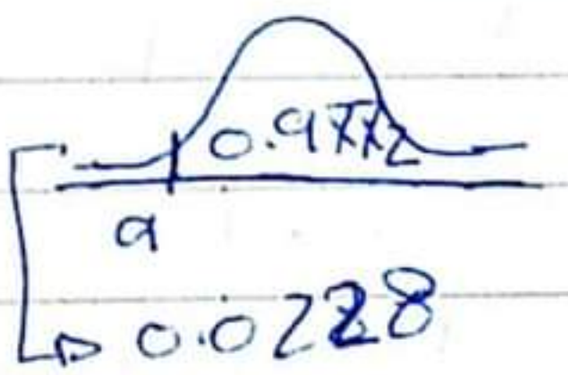
المساحة على اليسار $P(z < a) = 0.975$ ← بدور عليها

a إلى

$\leftarrow 0.025 + 0.95 = 0.975$ من الجدول

$a = 1.86$

Ex) ⑥ Find a s.t $P(z > a) = 0.9772$



$1 - P(z \leq a) = 0.9772$

$P(z \leq a) = \boxed{0.0228}$ ← من الجدول

$a = -2$

Ex-1

1.15

1.1

بدور عليها بالعمود

0.05

بدور عليها بالصف

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189

Ex-3

0.9

اقرب اشي 0.89937

الجواب: 1.28

العمود مع الصف

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409

0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

wed
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prob. CH.4

Ex) suppose that ranges of some signals are normally distributed with [mean 15 Kms] and [Std 3Kms]. → 6 (μ)

a) Find the proportion (النسبة) of the signals that exceeds (يتجاوز) 17 kms.

(صيغة ثانية للسؤال) If a signal is chosen at random, find the prob. that its range will exceed 17 kms

(المدى الذي يخطر ببالك) $X \sim N(15, 9)$

{proportion = prob}

(يكون المقام 6 ويقسم على 6)

$$P(X > 17) = P\left(\frac{X-\mu}{6} > \frac{17-\mu}{6}\right)$$

Std = 3
$6^2 = 9$

$$= P\left(Z > \frac{17-15}{3}\right)$$

(الجدول بتعامل فقط مع الحالة الكلاسية للـ Normal dist اللي هي Std Normal dist)

$$= P(Z > 0.6\bar{7})$$

$$= 1 - P(Z \leq 0.6\bar{7})$$

(عشان نقدر نلاقيها على الجدول بدنا القيمة على اليسار فأخدنا المتممة)

$$= 1 - T(0.6\bar{7})$$

$$= 1 - 0.74857$$

$$= 0.25143$$

(μ = 1 لـ $6^2 = 0$)

(وتكامل p.d.f هكينا ما بقدر أوجده حسابيًا بدنا نلجأ للجدول. فلازم نحول الـ Normal لـ Std Normal)

كيف؟ $Z = \frac{X-\mu}{6} \sim N(0,1)$

b) Find the proportion of signals with ranges less than 10 Kms.

$$P(X<10) = P\left(\frac{X-\mu}{6} < \frac{10-\mu}{6}\right)$$

$$= P\left(Z < \frac{10-15}{3}\right)$$

$$= P(Z < -1.67) = T(-1.67)$$
$$= 0.04746.$$

C) Find the proportion of signals with ranges between 10 and 17 Kms.

$$P(10<X<17) = P\left(\frac{10-\mu}{6} < \frac{X-\mu}{6} < \frac{17-\mu}{6}\right)$$

$$= P\left(\frac{10-15}{3} < Z < \frac{17-15}{3}\right)$$

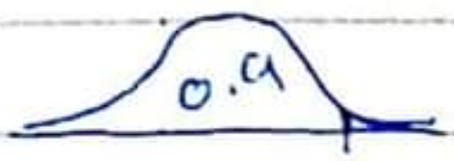
$$= P(-1.67 < Z < 0.67)$$

$$= T(0.67) - T(-1.67)$$
$$= 0.74857 - 0.04746 = 0.70111$$

d) Find the range of signal that are best than 90% of other signals.

المساحة على اليسار

$$\frac{90}{100} = 0.9$$



$$P(Z<a) = 90\%$$

prob. CM.4 — wed 8-Dec-2021

but $Z = \frac{X-\mu}{6}$ — لأن Z طلع X بده

$Z = 1.28$

$1.28 = \frac{X-15}{3} \Rightarrow X = 3(1.28) + 15 = 18.84$ kms

e) Find the range of signal that is considred with max range of the worst 5% of signals.

5%



بدور على 0.05 هوا الجدول بتطلع Z

المساحة على اليمين أكبر مدى للإشارة الذي يعبر أسوء ب 5% من كل الإشارات. أكبر الـ Range

$Z = -1.64$ ← من الجدول

$X = 6Z + \mu$ ← الـ Z اللي فوق من قانون

$(3)(-1.64) + 15 = 10.08$ kms.

f) $P(X = 16) = 0$, Cont r.V احتمال عند نقطة = 0 في Cont

g) If two signals are chosen at random, find the prob. that their ranges will exceed 17 kms.

$P(X_1 > 17 \cap X_2 > 17)$

indep ←

$P(X_1 > 17) \cdot P(X_2 > 17)$

$= (0.2514\mathbb{B})^2$

نفس نوع ← $P(X_1 > 17) = 0.25143$

$P(X_2 > 17) = 0.25143$

h) If 5 signals are chosen at random, find the prob. that the ranges of two of them will exceed 17 Kms.

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prob. CH.4 wed 8.Dec.2021

5 signals ← بدي اختار 2

$P(X_1 > 17) = 0.25143$ حتى كل محاولة من 5 محاولات احتمال النجاح

$P(X_2 > 17) = 0.25143$

$\vdots$

$P(X_5 > 17) = 0.25143$ وبدي

All signals are indep. and Identical ← Binomial

$n = 5$

$Y \sim B(5, 0.25143)$

$P(Y=2) = \binom{5}{2} (0.25143)^2 (1-0.25143)^{5-2}$

$= 0.26517$

$\frac{5!}{2!3!}$

$= \frac{5 \times 4 \times 3!}{2 \times 1 \times 3!} = \boxed{10}$

prob. CH.4

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Transformation of random variable:

If $f_X(x)$ is the p.d.f of a r.v X and

new r.v $Y = T(X)$ is a function of the r.v X,

we want to find the distribution of a r.v Y.

بدالة X ← تبدي p.d.f للمتغير العشوائي الجديد.

$$F_Y(y_0) = P\{Y \geq y_0\} = P\{T(X) \geq y_0\}$$

$$= P\{X \leq T^{-1}(y_0)\}$$

$$= F_X(x)$$

$$f_Y(y) = f_X(T^{-1}(y)) \left| \frac{d\,T^{-1}(y)}{dy} \right| \leftarrow \text{Jacobian}$$

T inverse y → X

Ex) Let $y = aX + b$, then

① X موضوع القانون

② نشتق $\frac{dx}{dy} = \frac{1}{a}$ ← $x = \frac{y-b}{a}$

③ نعوض بالقانون

$$f_Y(y) = f_X\left(\frac{y-b}{a}\right) \left| \frac{1}{a} \right|$$

Ex) If $y = aX^2 + b$, then مشتقة ما بداخل الجذر

$$x = \pm\sqrt{\frac{y-b}{a}} \rightarrow \frac{dx}{dy} = \frac{\frac{1}{a}}{\pm 2\sqrt{\frac{y-b}{a}}}$$

$$f_Y(y) = f_X\left(\pm\sqrt{\frac{y-b}{a}}\right) \left| \frac{\frac{1}{a}}{\pm 2\sqrt{\frac{y-b}{a}}} \right|$$

↳ 2* اكثر نفسه

prob. CH.4 — Mon 13.Dec.2021

Ex) Consider the dist, with p.d.f

p.d.f ←

$$f(x) = \frac{\theta^2}{\theta+1}(1+x)e^{-\theta x}, \quad \boxed{x>0}$$

, Find the desity of y s.t.

p.d.f ←

the same.

p.d.f density distribution

$$y = e^x - 1, \quad \boxed{y>0}$$

ما بنا داعي للمطلق

$$y = e^x - 1 \rightarrow x = \ln(y+1)$$

$$\frac{dx}{dy} = \frac{1}{y+1}$$

مشتقة ما داخل ln ← ln ← ما داخل ln نفسه

$$f_Y(y) = f_X(\ln(y+1)) \left|\frac{1}{y+1}\right|$$

$$= \frac{\theta^2}{\theta+1}\left(1+\ln(y+1)\right) e^{-\theta(\ln(y+1))} \left|\frac{1}{y+1}\right|$$

$$= \frac{\theta^2}{\theta+1}\left(1+\ln(y+1)\right)(y+1)^{-\theta}\frac{1}{y+1}$$

$$= \frac{\theta^2}{\theta+1}\left(1+\ln(y+1)\right)\frac{1}{(y+1)^{\theta+1}} \quad \frac{1}{(y+1)^{\theta}} * \frac{1}{(y+1)^{1}}$$

$$y = \frac{1}{x} \Rightarrow x = \frac{1}{y} \rightarrow \frac{dx}{dy} = \frac{-1}{y^2}$$

$$f_Y(y) = f_X\left(\frac{1}{y}\right)\left|\frac{-1}{y^2}\right|$$

$$= f_X\left(\frac{1}{y}\right)\frac{1}{y^2}$$

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Def. The correlation of x and Y is defined by $R_{xy} = E(xy)$

Remarks : [1] If x and y are independent then $E(xy) = E(x) \cdot E(y)$

[2] If $R_{xy} = 0$, then x and y are orthogonal.

Example - Let x be a r.v with $E(x) = 3$, $6x^2 = 2$ and $y = -6x + 22$

[1] Find the correlation of x, y.

[2] Are x and y independent?

[3] Are x, y correlated?

Sol : [1] $R_{xy} = E_{xy} = E(x(-6x+22))$

$= E(-6x^2 + 22x)$

$= -6E(x^2) + 22E(x)$

since $6x^2 = 3$ (variance) $\rightarrow 6x^2 = E(x^2) - (E(x))^2$

$2 = E(x^2) - 3^2$

$E(x^2) = 11$

$R_{xy} = -6E(x^2) + 22E(x)$

$= -6(11) + 22(3)$

$R_{xy} = 0 \rightsquigarrow$ x, y orthognal

[2] $E(y) = E(-6x + 22) = -6E(x) + 22$

$= -6(3) + 22 = 4$

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$E(X) \cdot E(Y) = (3)(4) = 12 \neq E(XY) = 0$

X, Y are dependent.

[3] since X, Y are dependent, then X, Y are correlated.

Remark :-

[1]. If X, Y are dependent $\Rightarrow$ X, Y are correlated

[2]. If X, Y are independent $\Rightarrow$ X, Y are uncorrelated

* $Cov(X, Y) = C_{XY}$ ⇓

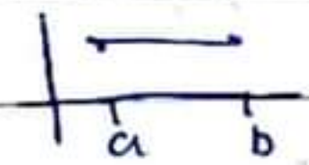
$$corr(X, Y) = \frac{C_{XY}}{\sigma_X \cdot \sigma_Y}$$

[4] The uniform distribution.

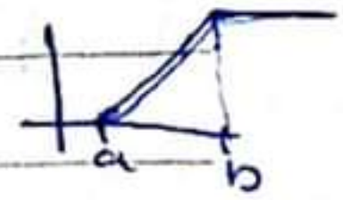
Def: The uniform r.v X is denoted by

$X \sim U(a, b)$, with p.d.f of

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$



and c.d.f of $F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$



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Random process r.p

Def: The random process "r.p" or (stochastic process). is an infinite indexed family $(set)_{\Omega}$

$X(t,s) = \{x_j(t,s)\}$, such that:

$t \rightarrow$ time

$s \in \Omega$: is a possible outcomes of an experiment.

$t \rightarrow$ يوجد اقتران بمتغير

$s \rightarrow$ fixed = ثابت

Remarks: ① $X(t,s_0)$ is called [Sample function] or (ensemble function) or (realization of a process).

fixed ثابت

② $X(t_0,s)$ is a random variable on Ω.

بتجمد الزمن بقيمة ثابتة t_0

③ for frozen time $t = t_1$, then:

$X_1 = X(t_1,s) = X(t_1)$ describes statistically the properties of the r.v at time t_1 ثابت.

④ $E(X_1)$: mean value of the process which is called ensemble average.

⑤ In general:

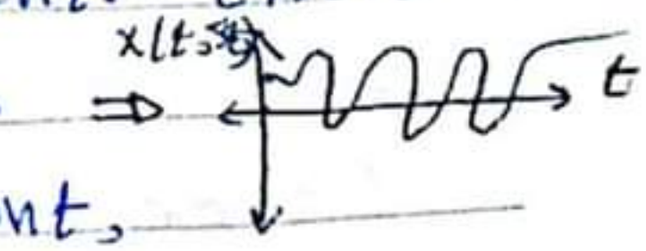
$X_i = X(t_i, s) = X(t_i)$ $\quad i = 1,2,3,\dots$

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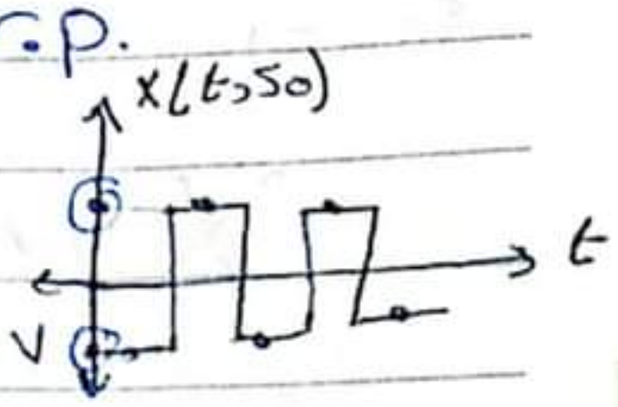
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* Classification of process:

① If X is continuous and t is cont. then the X(t) is called a cont r.p. ⇒



② If X is discrete and t is cont, then X(t) is called a discrete r.p.

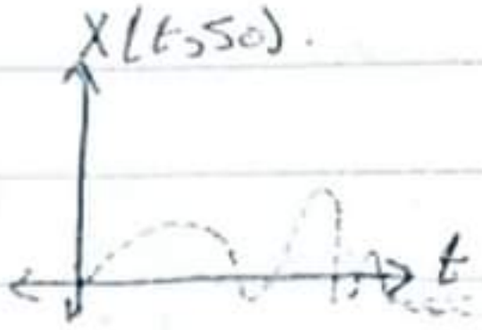


Note: The sample function has only two values.

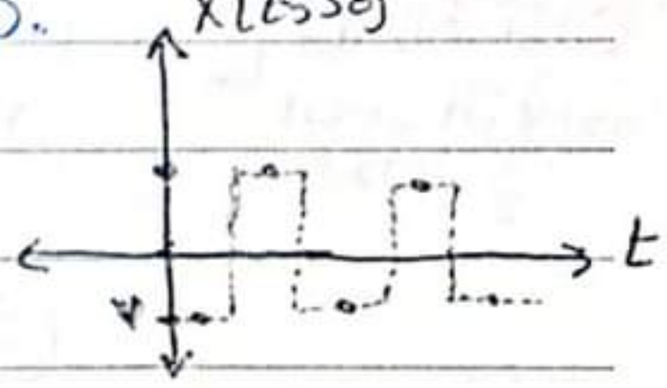
③ If X is Cont and t is discrete, then X(t) is Cont r.p.

Note: D.T ⇐ Discrete time

C.T ⇐ Cont time.



④ If X is discrete and t is discrete, then X(t) is called a discrete r.p.



Remark: In a deterministic r.p. any future values of any [sample function] can be [predicted] from the past ones.

[⇐ X(t,so). ⇐ التوقع]

Ex = X(t) = A cos(ωt + Θ), when

[ثوابت ⇐] [⇐ بكون محدد من السؤال ايش التوزيع اللي عليه]

A & ω: are fixed

Θ: r.v [which is determined]

X(t) as a function of time is entirely specified.

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- Remember that : Two Random variables X and Y are independent if : $f(X,Y) = f_X(X) \cdot f_Y(y)$.
- Remark : Two Random process $X(t)$, $y(t)$ are statistically independent if:

$$f(X_1, X_2, \dots X_n, y_1, y_2, \dots y_n, t_1, t_2, \dots t_n, t'_1, t'_2, \dots t'_n)$$
$$= f_x(X_1, X_2, \dots X_n, t_1, t_2, \dots t_n) * f_y(y_1, y_2, \dots y_n, t'_1, t'_2, \dots t'_n)$$

$$X(t) = \{ X(t,s) \} = \{ X_1(t_1), X_2(t_2), \dots, X_n(t_n) \}$$

$$y(t) = \{ y(t,s) \} = \{ y_1(t'_1), y_2(t'_2), \dots, y_n(t'_n) \}$$

has two types : 1-first-order , 2-second-order.

Def → The random process is called stationary if all of it's statistical properties don't change with time.

Def → A random process is called first-order stationary if it's first-order density function doesn't change with time shift i.e. →

$$f_x(x, t_1) = f_x(x, t_1 \pm \Delta)$$

This means the first-order density function of X $f_x(x, t_1)$ is independent. so,

$$E(X(t)) = \bar{X} = \text{Constant}.$$

Def → A random process is called second-order stationary if it's second-order density function satisfies:

$$f_x(X_1, X_2, t_1, t_2) = f_x(X_1, X_2, t_1 \pm \Delta, t_2 \pm \Delta)$$

- Recall: $R_{xy} = E(XY)$ = The correlation of X and Y (for random variable).

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Def:- The auto-correlation function of the random process $X(t)$ is defined by:

$$R_{XX}(t_1, t_2) = E(X(t_1) \cdot X(t_2)) = E(X_1, X_2)$$

Let $t_1 = t$ and $t_2 = t_1 + \tau \Rightarrow \tau = t_2 - t_1$

$$R_{XX}(t, t+\tau) = E(X(t) \cdot X(t+\tau))$$

Remark: The r.p $X(t)$ is called second-order stationary if the auto-correlation function of $X(t)$ satisfies:

$$R_{XX}(t, t+\tau) = R_{XX}(\tau)$$

Stationary

- first-order stationary if: $E(X(t)) = \text{constant}$
- second-order stationary if: $R_{XX}(t, t+\tau) = R_{XX}(\tau)$
- ⊕ if both conditions are satisfied

The Random process is wide-sense stationary.

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Ex- Show that the process $X(t) = A\cos(\omega_0 t + \theta)$ is wide-sense stationary if A, ω_0 constant and θ is uniformly distribution random variable on $(0, 2\pi)$

Sol:

$$\theta \sim U(0, 2\pi) \Rightarrow f(\theta) = \begin{cases} 1/2\pi & , 0 \le \theta \le 2\pi \\ 0 & , \text{otherwise} \end{cases}$$

for it to be wide-sense stationary, we have to prove that

[1] $E(X(t)) =$ constant [2] $R_{XX}(t, t+\tau) = R_{XX}(\tau)$

[1] $E(X(t)) = E(A\cos(\omega_0 t + \theta))$

$= A\,E(\cos(\omega_0 t + \theta)) = A\int_0^{2\pi} \cos(\omega_0 t + \theta) f(\theta)\, d\theta$

$= A\int_0^{2\pi} \cos(\omega_0 t + \theta)\, 1/2\pi\, d\theta$

$= A/2\pi \left[\sin(\omega_0 t + \theta)\right]_0^{2\pi}$

$= A/2\pi \left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t)\right]$

$= A/2\pi \left[\sin(\omega_0 t)\cos(-2\pi) + \sin(2\pi)\cos(\omega_0 t) - \sin(\omega_0 t)\right]$

(1) Zero

$= A/2\pi \left[\sin(\omega_0 t) - \sin(\omega_0 t)\right] = \text{Zero}$

constant

$\therefore E(X(t)) =$ Constant $= 0$

$X(t)$ is first-order stationary

[2] $R_{XX}(t, t+\tau) = E(X(t) \cdot X(t+\tau))$

$= E(A\cos(\omega_0 t + \theta) \cdot A\cos(\omega_0 t + \tau) + \theta)$

$= A^2 E(\cos(\omega_0 t + \theta) \cdot \cos(\omega_0 t + \omega_0 t + \theta)$

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- Remember:

⊕ $\cos(a+b) = \cos a \cos b - \sin a \sin b$

$\cos(a-b) = \cos a \cos b + \sin a \sin b$ e

$= \cos(a)\cos(b) = 1/2\,(\cos(a+b) + \cos(a-b))$.

$L_D\ A^2/2\ E(\cos(2w_0t + w_0\tau + 2\theta)) + \cos(-w_0\tau)$

$= A^2/2\ E(\cos(2w_0t + w_0\tau + \tau + 2\theta) + \cos(w_0\tau))$

$= A^2/2\ E(\cos(2w_0t + w_0\tau + 2\theta)) + E(\cos(w_0\tau))$ → $E(a)=a$

$= A^2/2 \left(\int_0^{2\pi} \cos(2w_0t + w_0\tau + 2\theta)\,\frac{1}{2\pi}\,d\theta\right) + \cos(w_0\tau)$.

$= A^2/2\,(1/2\pi\ \tfrac{1}{2} \sin(2w_0t + w_0\tau + 2\theta)]_0^{2\pi} + \cos(w_0\tau)$.

$= A^2/2\,(1/4\pi \sin(\underbrace{2w_0t + w_0\tau}_{X} + \underbrace{4\pi}_{Y}) - 1/4\pi[\sin(2w_0t + w_0\tau)] + \cos(w_0t)$.

Since: $\sin(\overbrace{2w_0t + w_0\tau}^{x} + \overbrace{4\pi}^{y})$

$= \sin(2w_0t + w_0\tau)\cos(4\pi)^{(1)} + \cos(2w_0t + w_0\tau)\sin(4\pi)$ (zero)

$= \sin(2w_0t + w_0\tau)$

$\therefore R_{XX}(t, t+\tau) = A^2/2\,(1/4\pi \sin(2w_0t + w_0\tau) - 1/4\pi \sin(2w_0t + w_0\tau + \cos(w_0\tau))$.

$R_{XX}(t, t+\tau) = A^2/2 \cos(w_0\tau)$

$= R_{XX}(\tau)$.

→ $X(t)$ is second-order stationary.

- Remark: for any r.p, the auto-correlation function statifies the following:

1) $|R_{XX}(\tau)| \leq R_{XX}(0)$. $|x| \leq a \rightarrow -a \leq x \leq a$

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[2] $R_{XX}(-\tau) = R_{XX}(\tau)$

[3] $R_{XX}(0) = R_{XX}(t, t+0)$
$= E(X(t) \cdot X(t))$
$= E(X^2(t))$

[4] $\lim_{|\tau| \to \infty} R_{XX}(\tau) = \overline{X}^2$

*Recall: $\overline{X} = E(X(t))$ the $\overline{X}^2 = (E(X(t)))^2$

[5] If $X(t)$ is periodic, then $R_{XX}(\tau)$ is perf
with the same period.

Def: A process $X(t)$ with constant mean value $\overline{X}$ is called mean-ergodic $E(X(t)) = \overline{X}$

Example- Given the auto-correlation function for a stationary ergodic process $R_{XX}(\tau) = 25 + \frac{4}{1+6\tau^2}$ (second order stationary)
Find the mean and varience of the process θ.

Sol:

$$\lim_{\tau \to \infty} R_{XX}(\tau) = \lim_{\tau \to \infty} 25 + \frac{4}{1+6\tau^2} = 25 + \frac{4}{\infty} = 25$$

$$\overline{X}^2 = 25 \Rightarrow \overline{X} = \pm 5$$

$$E(X^2(t)) = R_{XX}(0) = 25 + \frac{4}{1+0} = 29$$

$$\sigma^2_{X(t)} = E(X^2(t)) - E((X(t))^2) = 29 - (\pm 5)^2 = 4$$

✱ Cross - Correlation functions:

Def: The cross-correlation function of two random processes is given by:

$$R_{xy}(t, t+\tau) = E(X(t) \cdot Y(t+\tau))$$

Remarks → 1) If $X(t)$ and $Y(t)$ are jointly wide-sense stationary, then:

$$R_{xy}(t, t+\tau) = R_{xy}(\tau)$$

2) If $R_{xy}(\tau) = 0$, then $X(t)$ and $Y(t)$ are orthogonal processes.

3) If $x(t)$ and $Y(t)$ are independent then

$$R_{xy}(t, t+\tau) = E(X(t)) \cdot E(Y(t+\tau))$$

4) If $X(t)$ and $Y(t)$ are independent and at least one of them is wide-sense stationary, then:

$$R_{xy}(\tau) = \overline{X} \cdot \overline{Y} = \text{Constant}$$

5) $R_{xy}(-\tau) = R_{xy}(\tau)$

6) $|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0) \cdot R_{yy}(0)}$ أكبر قيمة

7) $|R_{xy}(\tau)| \leq 1/2 (R_{xx}(0) + R_{yy}(0))$.

Example - Assume that the two random processes:

$$X(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$$

$$Y(t) = B\cos(\omega_0 t) - A\sin(\omega_0 t)$$

where ω_0 is Constant, A & B are uncorrelated ← independent random variables with zero-mean and have the same variance.

Find:

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[1] The cross-correlation function
[2] show that X(t) and Y(t) are jointly wide sense stationary. Remark (1)

Sol:

Since A and B are uncorrelated
A & B are independent
$E(AB) = E(A) \cdot E(B) = 0$. $(E(A) = 0 , E(B) = 0)$ Zero-mean
$\sigma_A^2 = \sigma_B^2 \Rightarrow \sigma_A = \sigma_B = \sigma$ (same varience)

[1] $R_{XY}(t, t+\tau) = E(X(t) \cdot Y(t+\tau))$
$= (A\cos(\omega_0 t) + B\sin(\omega_0 t)) \cdot (B\cos(\omega_0 t + \omega_0 \tau) - A\sin(\omega_0 t + \omega_0 \tau))$
$= AB\cos(\omega_0 t)\cos(\omega_0 t + \omega_0 \tau) - A^2\cos(\omega_0 t)\sin(\omega_0 t + \omega_0 \tau)$
$+ B^2\sin(\omega_0 t)\cos(\omega_0 t + \omega_0 \tau) - AB\sin(\omega_0 t)\sin(\omega_0 t + \omega_0 \tau)$

$R_{XY}(t, t+\tau) = \overbrace{E(AB)}^{\text{zero}}\, E(\cos(\omega_0 t)\cdot\cos(\omega_0 t + \omega_0 \tau)) - E(A^2)\, E(\cos(\omega_0 t)\cdot\sin(\omega_0 t + \omega_0 \tau))$
$+ E(B^2)\, E(\sin(\omega_0 t)\cos(\omega_0 t + \omega_0 \tau)) - \overbrace{E(AB)}^{\text{zero}}\, E(\sin(\omega_0 t)\cdot\sin(\omega_0 t + \omega_0 \tau))$

توقع الثابت = ثابت

$= -\sigma^2(\cos(\omega_0 t)\sin(\omega_0 t + \omega_0 \tau) - \sin(\omega_0 t)\cos(\omega_0 t + \omega_0 \tau))$

$\sigma_A^2 = E(A^2) - \overbrace{(E(A))^2}^{0}$
$\sigma_B^2 = E(B^2) - \overbrace{(E(B))^2}^{\text{zero}}$
$\Rightarrow E(A^2) = E(B^2) = \sigma^2$

* Recall ⇒
$\sin(a-b) = \sin a\cos b - \cos a\sin b$
$a = \omega_0 t + \omega_0 \tau$, $b = \omega_0 t$

$R_{XY}(t, t+\tau) = -\sigma^2\sin(\omega_0 t + \omega_0 \tau - \omega_0 t)$
$= -\sigma^2\sin(\omega_0 \tau)$ ← دالة في τ

So: $R_{XY}(t, t+\tau) = R_{XY}(\tau)$
⇒ X(t) and Y(t) are jointly wide-sense stationary

Ex- Let $X(t)$ be a wide-sense stationary with auto-correlation function $R_{XX}(\tau) = e^{-a|\tau|}$. Let $Y(t) = X(t)\cos(\omega_0 t + \theta)$, where ω_0 is constant and θ is uniform r.v on $(-\pi, \pi)$, $\cos(\omega_0 t + \theta)$ is independent from $X(t)$". Find the auto-correlation function of $Y(t)$.

Sol: $R_{YY}(t, t+\tau) = E(Y(t) \cdot Y(t+\tau))$

$= E(X(t)\cdot\cos(\omega_0 t+\theta)\cdot X(t+\tau)\cos(\omega_0 t+\omega_0\tau+\theta))$

$= E(X(t)X(t+\tau)) \cdot E(\cos(\omega_0 t)\cos(\omega_0 t+\omega_0\tau+\theta))$'''

$= R_{XX}(t, t+\tau) \cdot \int_{-\pi}^{\pi} \cos(\omega_0 t)\cos(\omega_0 t+\omega_0\tau+\theta)\, 1/2\pi\, d\theta$

$= R_{XX}(\tau) \int_{-\pi}^{\pi} 1/4\pi\,(\cos(2\omega_0 t+\omega_0\tau+2\theta) + \cos(-\omega_0\tau))\, d\theta$

$f(\theta) = \begin{cases} \frac{1}{2\pi}, & -\pi < \theta \le \pi \\ 0, & \text{otherwise} \end{cases}$

* Recall - $\cos(a+b) = \cos a\cos b - \sin a \sin b$

(+) $\cos(a-b) = \cos a\cos b + \sin a \sin b$

$= \cos a\cos b = 1/2(\cos(a+b) + \cos(a-b))$

$= e^{-a|\tau|} \cdot 1/4\pi \left(\left[1/2 \sin(2\omega_0 t+\omega_0\tau+2\theta) \right]_{-\pi}^{\pi} + \left[\cos(\omega_0 t)\cdot\theta \right]_{-\pi}^{\pi} \right)$

$= e^{-a|\tau|} \cdot 1/4\pi\,(0 + 2\pi\cos(\omega_0\tau))$

$= 1/2\, e^{-a|\tau|}\cos(\omega_0\tau)$

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Def: ① The auto-covariance function of the random process $X(t)$ is defined by:

$$C_{XX}(t, t+\tau) = E(X(t) \cdot X(t+\tau)) - E(X(t)) \cdot E(X(t+\tau))$$
$$= \underbrace{R_{XX}(t, t+\tau)}_{\text{auto-correlation fun}} - E(X(t)) \cdot E(X(t+\tau))$$

② The cross covariance function of the random processes $X(t)$ and $y(t)$ is defined by:

$$C_{XY}(t, t+\tau) = E(X(t) \cdot y(t+\tau)) - E(X(t)) E(Y(t+\tau))$$
$$= \underbrace{R_{XY}(t, t+\tau)}_{\text{cross correlation function}} - E(X(t)) E(Y(t+\tau))$$

*Remarks:

① If $X(t)$ and $Y(t)$ are wide-sense stationary, then:

Ⓐ $C_{XX}(t, t+\tau) = R_{XX}(\tau) - \underbrace{\overline{X}^2}_{\text{constant}}$

Ⓑ $C_{XY}(t, t+\tau) = R_{XY}(t, t+\tau) - \underbrace{\overline{X} \cdot \overline{Y}}_{\text{constant}}$

② If $X(t)$ and $Y(t)$ are jointly wide-sense stationary, then:

Ⓐ $C_{XX}(t, t+\tau) = R_{XX}(\tau) - \overline{X}^2$

Ⓑ $C_{XY}(t, t+\tau) = R_{XY}(\tau) - \underbrace{\overline{X} \cdot \overline{Y}}_{\text{constant}}$

③ If $\tau = 0$, then:

$$C_{XX}(t, t+\tau)\Big|_{\tau=0} = E(X^2(t)) - (E(X(t)))^2 = Var(X(t))$$

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④ If $X(t)$, $Y(t)$ are independent random process, then:

$$C_{XY}(t, t+\tau) = 0$$

* Discrete-time random process

- It is defined as $X(t) = X(nT_s)$ where:

$n = 0, \pm 1, \pm 2, \pm 3, \ldots$

T_s = sampling interval.

- If $t_1 = n_1 T_s$ and $t_2 = n_2 T_s$ then we choose $\tau = t_2 - t_1$

$$= n_2 T_s - n_1 T_s = \underbrace{T_s (n_2 - n_1)}_{K} = KT_s, \text{ so:}$$

① The mean of the r.p $X(t)$

$$E(X(nT_s))$$

② Auto-correlation function is given by:

$$R_{XX}(nT_s, nT_s + KT_s) = E(X(nT_s) \cdot X(nT_s + KT_s))$$

$$- E(X(nT_s)) \cdot E(nT_s + KT_s)$$

③ cross-correlation function of $X(t)$ and $Y(t)$ is given by:

$$R_{XY}(nT_s, nT_s + KT_s) = E(X(nT_s) \cdot Y(nT_s + KT_s))$$

$$- E(X(nT_s)) \cdot E(Y(nT_s + KT_s))$$

④ The auto-covariance function of $X(t)$ is given by:

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$C_{XX}(nT_s, nT_s + kT_s) = R_{XX}(nT_s, nT_s + kT_s) - E(X(nT_s)) \cdot E(X(nT_s + kT_s))$

⑤. The cross-covariance function of $X(t)$ and $Y(t)$ is given by:

$C_{XY}(nT_s, nT_s + kT_s) = R_{XY}(nT_s, nT_s + kT_s) - E(X(nT_s)) \cdot E(Y(nT_s + kT_s))$

* Power density spectrum

- The spectral properties of a signal $X(t)$ are contained in its fouries transform $X(w)$ given by:

$$X(w) = \int_{-\infty}^{\infty} X(t) \cdot e^{-iwt} \, dt \quad \text{---} \quad (1)$$

+ $i = \sqrt{-1}$, t: Time , w: frequency

The function $X(w)$, sometimes called simply the spectrum of $X(t)$. has the unit of volts per hertz, where $X(t)$ is a voltage and describes the way in which relative signal voltage is distributed with frequency.

Remarks : ①If $X(w)$ is known, we can find $X(t)$ by inverse fouries transform:

$$X(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) \cdot e^{iwt} \, dw$$

prob.

Mon.
10. Jan. 2022

② $X(w)$ forms a complete description of $X(t)$. and vice versa.
الوكس صحيح

wed.
12. Jan. 2022

③ If we apply equation (XX1) to a random process, then $X(w)$ may not exist for most sample functions of the process.

④ From remark 3, For a random process: $X(t)$. Let $X_T(t)$ be defined as that portion of a sample function $X(t)$ that exists between $-T$ and T, then:

$$X_T(t) = \begin{cases} X(t) & , -T < t < T \\ 0 & , \text{otherwise} \end{cases}$$

by fourier series, then:

$$X_T(w) = \int_{-T}^{T} X_T(t)\, e^{-jwt}\, dt = \int_{-T}^{T} X(t)\, e^{-jwt}\, dt .$$

prob.

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Def: For the r.p $X(t)$, the power density spectrum $= \lim_{T\to\infty} \frac{E(X_T^2(\omega))}{2T}$

The power function $\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} E(X_T^2(\omega))\, d\omega$

frequency domain.

Def: For the random process $X(t)$, we defined (the power function) (the power formula) (the average power) as

$P_{XX} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} E(X^2(t))\, dt$

Time domain.

- Example - Consider the r.p $X(t) = A_0 \cos(\omega_0 t + \theta)$ where A_0 and ω_0 are constants and θ is a r.v uniformly distributed on $(0, \frac{\pi}{2})$ Find P_{XX}.

Sol: $f(\theta) = \begin{cases} \frac{2}{\pi}, & 0 \le \theta \le \frac{\pi}{2} \\ 0, & \text{otherwise} \end{cases}$

$E(X^2(t)) = E(A_0^2 \cos^2(\omega_0 t + \theta))$

$= A_0^2 E(\cos^2(\omega_0 t + \theta))$

$= A_0^2 \int_0^{\pi/2} \cos^2(\omega_0 t + \theta) \cdot \frac{2}{\pi}\, d\theta$

$= \frac{2A_0^2}{\pi} \int_0^{\pi/2} \frac{1}{2} + \frac{1}{2}\cos(2\omega_0 t + 2\theta)\, d\theta$

$= \frac{2A_0^2}{\pi} \left[\frac{1}{2}\theta + \frac{1}{4}\sin(2\omega_0 t + 2\theta) \right]_0^{\pi}$

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$$= \frac{2A_0^2}{\pi}\left[\frac{\pi}{4} + \frac{1}{4}\sin(2\omega_0 t + \pi) - \left(0 + \frac{1}{4}\sin(2\omega_0 t)\right)\right]$$

$$= \frac{2A_0^2}{\pi}\left(\frac{\pi}{4} + \frac{1}{4}\sin(2\omega_0 t)\cos(\pi) + \frac{1}{4}\sin\pi\cos(2\omega_0 t) - \frac{1}{4}\sin(2\omega_0 t)\right)$$

$$= \frac{2A_0^2}{\pi}\left(\frac{\pi}{4} - \frac{1}{2}\sin(2\omega_0 t)\right)$$

$$= \frac{A_0^2}{2} - \frac{A_0^2}{\pi}\sin(2\omega_0 t)$$

$$P_{XX} = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T}\left(\frac{A_0^2}{2} - \frac{A_0^2}{\pi}\sin(2\omega_0 t)\right)dt.$$

$$= \lim_{T\to\infty} \frac{1}{2T}\left[\frac{A_0^2}{2}(2T) - \left(\frac{A_0^2}{\pi}\cdot\frac{1}{2}\cos(2\omega_0 t)\right)\right]_{-T}^{T}$$

$$= \lim_{T\to\infty} \frac{1}{2T}\left(A_0^2 T + \frac{A_0^2}{2\pi\omega_0}\underbrace{\left(\cos(2\omega_0 T) - \cos(-2\omega_0 T)\right)}_{\text{zero}}\right)$$

$$= \lim_{T\to\infty} \frac{A_0 T}{2T}$$

$$= \frac{A_0^2}{2}$$

prob.

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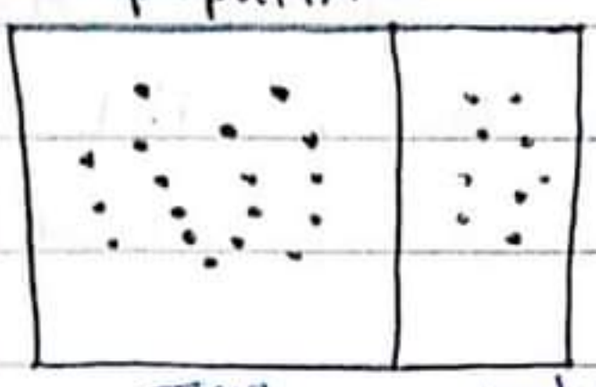
Mypothesis test: اختبار الفرضيات

دراسة لأعمار الرجال في مجتمع معين ← population.

parameters:
μ = mean of ages.
σ = std of ages

sampling ← تأخذ عينة

parameters:
$\bar{x}$ = mean of the sample.
s = std of the sample.



$\mu \simeq \bar{x}$ و
$\sigma \simeq s$

صحيح - لأن العينة كبيرة فتكون قريبة من الواقع.

نختبر إذا كان $\mu \simeq \bar{x}$ و $\sigma \simeq s$ ← لأن العينة صغيرة

Test ?! ← hypothesis test

هي طريقة لاتخاذ القرارات باستخدام البيانات سواء من تجربة خاضعة للسيطرة أم من دراسة غير خاضعة للتجربة.

Hypothesis test

[1] Ho : Null hypothesis
فرضية العدم
⇓
عكس Ha : تعني عدم وجود فرق بين العينة والـ population المأخوذ منها
أي تحقق :
$\mu \simeq \bar{X}$
$\sigma \simeq s$

[2] Alternative hypothesis
الفرضية البديلة
⇓

* إذا كان الوسط الحسابي للعينة التي يراد اختبارها هو $\mu_o = 50$ ،
فإن الخيارات الممكنة :

① $H_o : \mu = \mu_o$
$H_A : \mu \neq \mu_o$

② $H_o : \mu = \mu_o$
$H_A : \mu < \mu_o$

③ $H_o : \mu = \mu_o$
$H_A : \mu > \mu_o$

* عند وضع الفرضية واختبارها فإنه يوجد احتمالات الوقوع في الخطأ في اتخاذ القرارات :

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prob.

① Typ. I error (الخطأ من النوع الأول)

هو الخطأ الذي يقع فيه الباحث بعد قيامه بالعمل الإحصائي برفض الفرضية Ho، وقبول الفرضية Ha في الوقت الذي تكون فيه Ho صحيحة.

② Type II error (الخطأ من النوع الثاني)

هو الخطأ الذي يقع فيه الباحث به قيامه بالعمل الإحصائي بقبول الفرضية Ho، ورفض الفرضية Ha في الوقت الذي تكون فيه Ho خاطئة.

	True Ho	False Ho
Fail to reject Ho	Correct action	Type II error
Reject Ho	Type I error	Correct action